Test Sheet (Examination Database)

SECTION 1 — CONSTRAINTS TESTING

🔹 dbo.CK_Course_MinMaxDegree

Type: Constraint (Check)

📄 Description

Ensures course minimum degree is not greater than maximum degree.

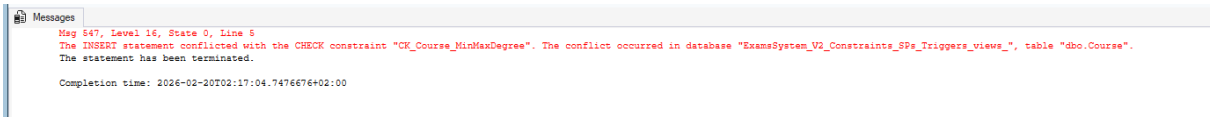
🧪 Test Query (Valid Insert)

```
INSERT INTO dbo.Course (CourseName, MaxDegree, MinDegree) V
ALUES ('Test', 100, 50);
```

🧪 Test Query (Invalid Insert)

```
INSERT INTO dbo.Course (CourseName, MaxDegree, MinDegree) V
ALUES ('Test', 50, 100);
```

📷 Test Result Screenshot



✅ Expected Result

- Valid insert succeeds.
- Invalid insert fails with check constraints.

🔎 Applied To which table

Course

🔹 Constraint: dbo.CK_Course_PositiveDegrees

Type: Foreign : Constraint (Check)

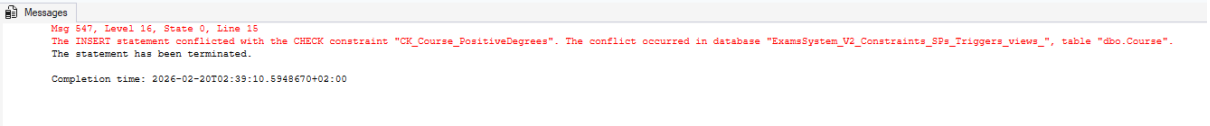
📄 Description

Ensures course degree limits are positive values.

🧪 Test Query

```
INSERT INTO dbo.Course (CourseName, MaxDegree, MinDegree) V
ALUES ('Test', -10, -20);
```

📷 Test Result Screenshot



✅ Expected Result

Insertion fails due to Negative Values.

🔎 Applied To which table

Course

🔹 Constraint: dbo.CK_CourseResult_FinalDegree

Type: Check Constraint

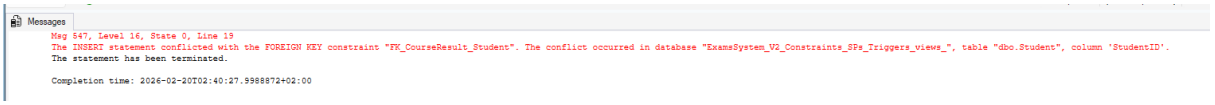
📄 Description

Ensures stored final degree is within valid score range.

🧪 Test Query

```
INSERT INTO dbo.CourseResult (FinalDegree, CourseID, Studen
tID) VALUES (999, 1, 1);
```

📷 Test Result Screenshot



✅ Expected Result

insert valid due to fact final degree is not in a valid score range

---

🔎 What This Proves

CourseResult

🔹 Constraint: dbo.CK_Exam_TimeValidation

Type: Check Constraint

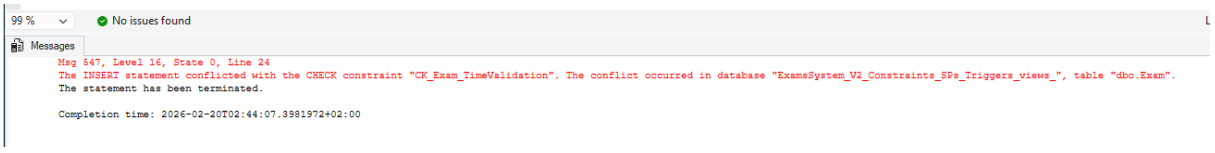
📄 Description

Ensures each exam start time is earlier than end time.

🧪 Test Query

```
INSERT INTO dbo.Exam (StartTime, EndTime , edate) VALUES
('14:00:00', '10:00:00' , '2020-05-05');
```

📷 Test Result Screenshot



✅ Expected Result

Inseration failed due to the fact that start time is after end time which breaks any logic.

🔎 Applied To which table

Exam

🔹 Constraint: dbo.CK_Exam_Type

Type: Check Constraint

📄 Description

Restricts exam type to supported values.

🧪 Test Query

```
INSERT INTO dbo.Exam (ExamType , edate ) VALUES ('InvalidTy
pe' ,'2022-05-05' );
```

📷 Test Result Screenshot

✅ Expected Result

Inseration failed because the exam type inserted in the list of exam tpye

---

🔎 Applied To which table

Exam

🔹 Constraint: dbo.CK_Question_Type

Type: Check Constraint

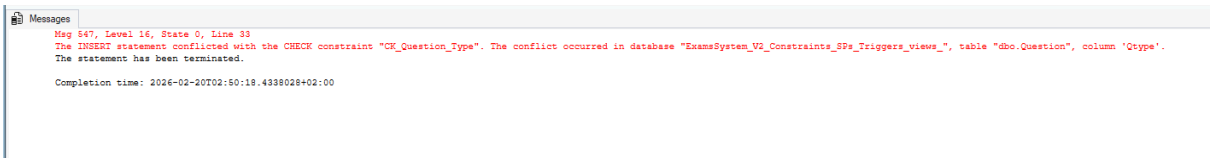
📄 Description

Restricts question type to supported values.

🧪 Test Query

```
INSERT INTO dbo.Question (Qtype) VALUES ('Unknown');
```

📷 Test Result Screenshot



✅ Expected Result

Failed due to fact that this type of question is not supported in the sytem

🔎 Applied To which table

Question

🔹 Constraint: dbo.CK_StudentExam_score

Type: Check Constraint

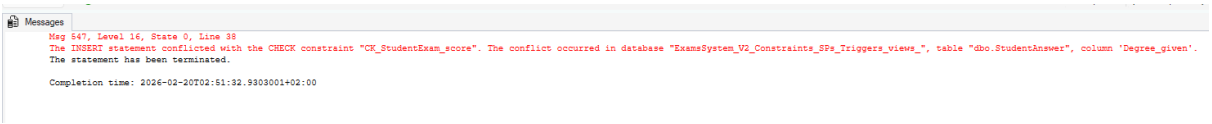
📄 Description

Ensures awarded answer score is within valid bounds.

🧪 Test Query

```
INSERT INTO dbo.StudentAnswer (Degree_given) VALUES (-5);
```

📷 Test Result Screenshot



✅ Expected Result

Inseration failed because the given answer score is not in range

🔎 Applied To which table

StudentAnswer

🔹 Constraint: dbo.CK_User_Role

Type: Check Constraint

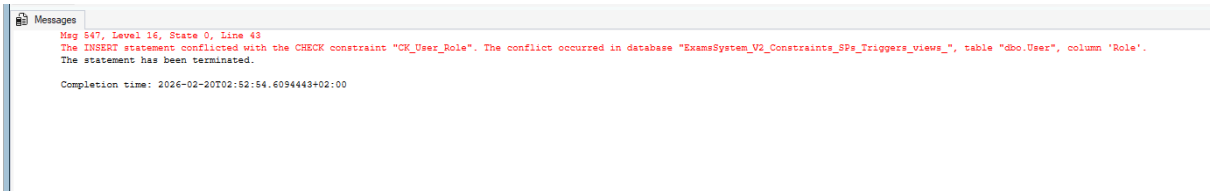
📄 Description

Restricts user role to supported system roles.

🧪 Test Query

```
INSERT INTO dbo.[User] (Username, [Role] )  VALUES ('testus
er', 'CEO');
```

📷 Test Result Screenshot



✅ Expected Result

Insert failed because the inserted role is not the list of valid roles in this column

🔎 Applied To which table

User

🔹 Constraint: dbo.UQ_OneAnswerPerQuestion

Type: Constraint (Unique)

📄 Description

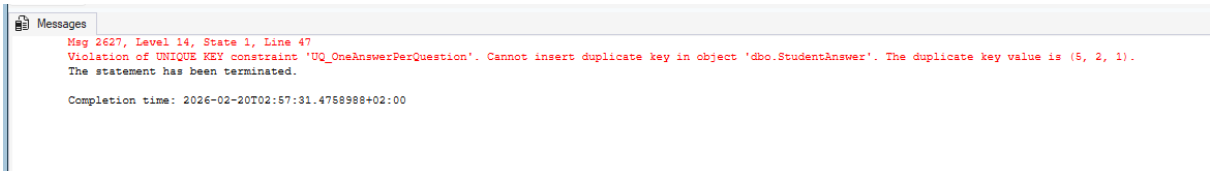
Prevents duplicate answers for the same student/exam/question. good for checking data redundancy Problems

---

🧪 Test Query

```
INSERT INTO dbo.StudentAnswer (STID, ExamID, QID, Answertex
t)
VALUES (5, 2, 1, 'TRUNCATE');
```

📷 Test Result Screenshot



✅ Expected Result

Insertion failed because this data already exists in the column

---

🔎 Applied To which table

StudentAnswer

🔹 Constraint: dbo.UQ_User_Username

Type: Check Constraint

📄 Description

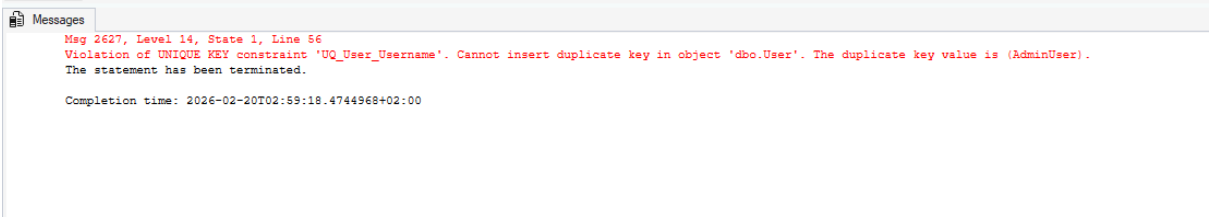
Restricts user role to supported system roles.

🧪 Test Query

```
-- Step 1: Insert first user
INSERT INTO dbo.[User] (Username, Role) VALUES ('AdminUse
r', 'Admin');

-- Step 2: Attempt duplicate username (This should FAIL)
INSERT INTO dbo.[User] (Username, Role) VALUES ('AdminUse
r', 'Student');
```

📷 Test Result Screenshot



✅ Expected Result

Couldn't Add new username because this username already exists in the table

---

🔎 Applied To which table

User

🧩 SECTION 2 — STORED PROCEDURES TESTING

Stored Procedure: dbo.sp_AddCourseToTrack

Type: Management Procedure

📄 Description

Adds a new course and associates it with a training track.

🧾 Parameters

Parameter	Data Type	Description
@CourseName	VARCHAR(50)	Name of the course
@Desc	VARCHAR(120)	Course description
@MaxDegree	INT	Maximum course degree
@MinDegree	INT	Minimum passing degree

🧪 Test Query

```
EXEC dbo.sp_AddCourseToTrack
    @CourseName='SQL Advanced',
    @Desc='Advanced SQL concepts',
    @MaxDegree=100,
    @MinDegree=60;
```

📷 Test Result Screenshot

```sql
--                    Testing Stored Procedur

EXEC dbo.sp_AddCourseToTrack
    @CourseName='SQL Advanced',
    @Desc='Advanced SQL concepts',
    @MaxDegree=100,
    @MinDegree=60;
```

99 % No issues found Ln: 61, Ch: 1

Messages

```
(1 row affected)

Completion time: 2026-02-20T03:24:36.6579592+02:00
```

✅ Expected Result

New course created and linked successfully.

🔎 Applied To Which Tables

- Course
- Track

2️⃣ Stored Procedure: dbo.sp_AddIntake

Type: Management Procedure

📄 Description

Creates a new intake period for a specific branch.

🧾 Parameters

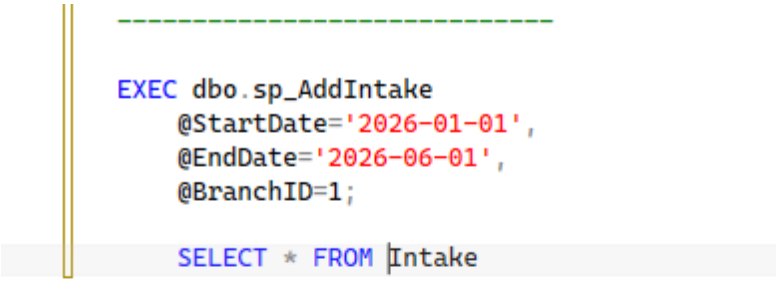
Parameter	Data Type
@StartDate	DATE

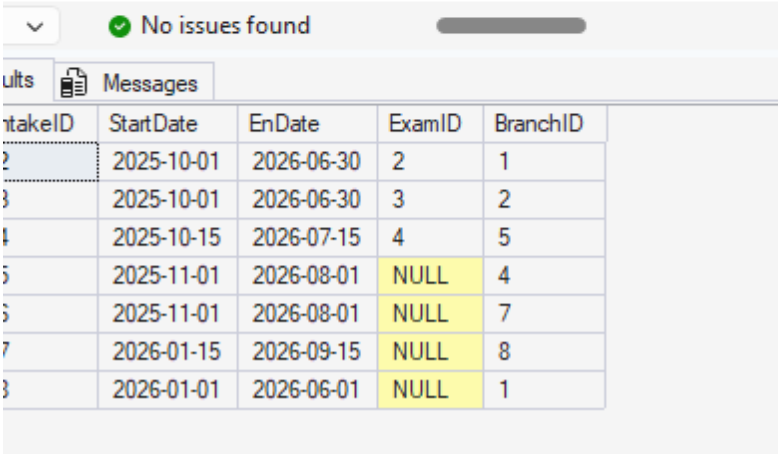
Parameter	Data Type
@EndDate	DATE
@BranchID	INT

🧪 Test Query

```
EXEC dbo.sp_AddIntake
    @StartDate='2026-01-01',
    @EndDate='2026-06-01',
    @BranchID=1;
```

📷 Test Result Screenshot





ntakeID	StartDate	EnDate	ExamID	BranchID
2	2025-10-01	2026-06-30	2	1
3	2025-10-01	2026-06-30	3	2
4	2025-10-15	2026-07-15	4	5
5	2025-11-01	2026-08-01	NULL	4
6	2025-11-01	2026-08-01	NULL	7
7	2026-01-15	2026-09-15	NULL	8
8	2026-01-01	2026-06-01	NULL	1

✅ Expected Result

New intake record inserted.

🔎 Applied To Which Tables

- Intake

3️⃣ Stored Procedure: dbo.sp_AddQuestion

Type: Instructor Procedure

📄 Description

Adds a new question to the question bank with type-specific details.

🧾 Parameters

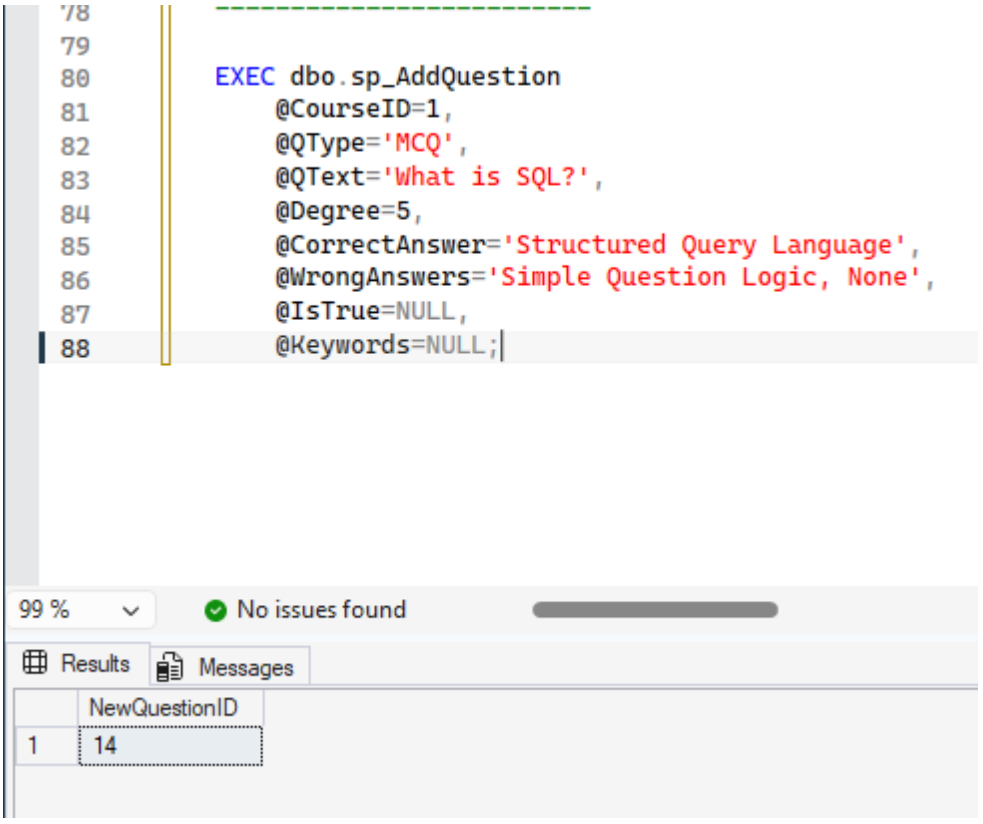
Parameter	Data Type
@CourseID	INT
@QType	VARCHAR(50)
@QText	NVARCHAR(MAX)
@Degree	INT
@CorrectAnswer	NVARCHAR(MAX)
@WrongAnswers	NVARCHAR(MAX)
@IsTrue	BIT
@Keywords	NVARCHAR(MAX)

🧪 Test Query

```
EXEC dbo.sp_AddQuestion
    @CourseID=1,
    @QType='MCQ',
    @QText='What is SQL?',
    @Degree=5,
    @CorrectAnswer='Structured Query Language',
    @WrongAnswers='Simple Question Logic, None',
```

```
    @IsTrue=NULL,
    @Keywords=NULL;
```

📷 Test Result Screenshot



✅ Expected Result

Question inserted and linked to course.

🔎 Applied To Which Tables

- Question
- MCQ / TrueFalseQuestion / TextQuestionKeywords
- CourseQuestion

Stored Procedure: dbo.sp_AssignStudentsToExam

Type: Instructor / Manager Procedure

📄 Description

Assigns eligible students to a specific exam.

🧾 Parameters

Parameter	Data Type
@ExamID	INT
@StudentID	INT
@TrackID	INT

🧪 Test Query

```
EXEC dbo.sp_AssignStudentsToExam
    @ExamID=1,
    @StudentID=10,
    @TrackID=2;
```

✅ Expected Result

Student successfully registered for the exam.

🔎 Applied To Which Tables

- StudentExam
- Student

5️⃣ Stored Procedure: dbo.sp_CreateRandomExam

Type: Business Logic Procedure

📄 Description

Creates an exam and auto-selects random questions by type.

🧾 Parameters

Parameter	Data Type
@CourseID	INT
@ClassID	INT
@StartTime	TIME
@EndTime	TIME
@ExamType	VARCHAR(30)
@Allowance	VARCHAR(75)
@ExamDate	DATE
@NumMCQ	INT
@NumTF	INT
@NumText	INT

🧪 Test Query

```
EXEC dbo.sp_CreateRandomExam
    @CourseID=1,
    @ClassID=1,
    @StartTime='10:00',
    @EndTime='12:00',
    @ExamType='Final',
    @Allowance='2 Hours',
    @ExamDate='2026-06-01',
    @NumMCQ=5,
    @NumTF=3,
    @NumText=2;
```

📷 Test Result Screenshot

```sql
EXEC dbo.sp_CreateRandomExam
    @CourseID=1,
    @ClassID=1,
    @StartTime='10:00',
    @EndTime='12:00',
    @ExamType='Normal',
    @Allowance='2 Hours',
    @ExamDate='2026-06-01',
    @NumMCQ=5,
    @NumTF=3,
    @NumText=2;
```

99 % No issues found

Results | Messages

	NewExamID	Message
1	14	Exam Created Successfully

🔎 Applied To Which Tables

- Exam
- ExamQuestion

6️⃣ Stored Procedure: dbo.sp_DeleteBranch_Hard

Type: Management Procedure

📄 Description

Permanently deletes a branch record after confirmation.

Prevents accidental deletion using a confirmation flag.

🧾 Parameters

Parameter	Data Type
@BranchID	INT
@ConfirmDelete	BIT

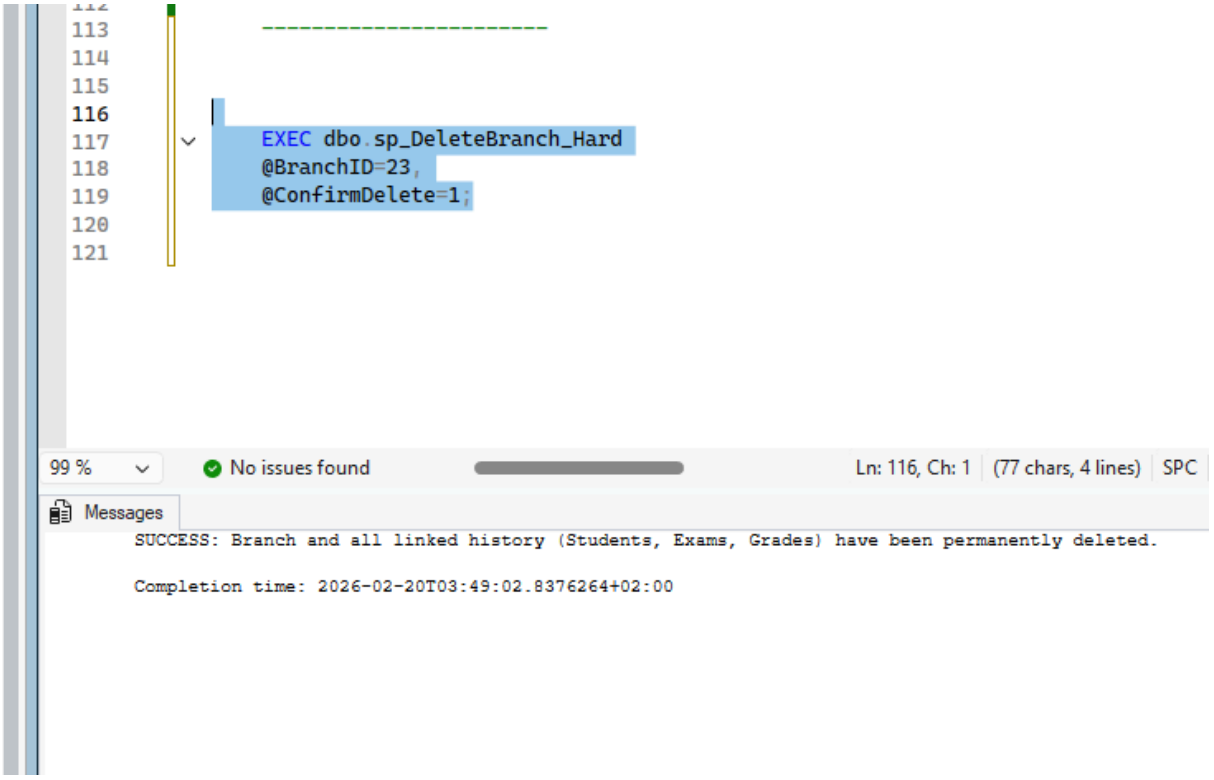
🧪 Test Query

```
EXEC dbo.sp_DeleteBranch_Hard
    @BranchID=23,
    @ConfirmDelete=1;
```

✅ Expected Result

Branch is deleted only if confirmation = 1.

📷 Test Result Screenshot



🔎 Applied To Which Tables

- Branch

7️⃣ dbo.sp_DeleteClass_Hard

📄 Description

Deletes a class permanently after confirmation.

🧾 Parameters

Parameter	Data Type
@ClassID	INT
@ConfirmDelete	BIT

🧪 Test Query

```
EXEC dbo.sp_DeleteClass_Hard
    @ClassID=5,
    @ConfirmDelete=1;
```

✅ Expected Result

Class record removed.

🔎 Applied To

- Class

8️⃣ dbo.sp_DeleteExam

📄 Description

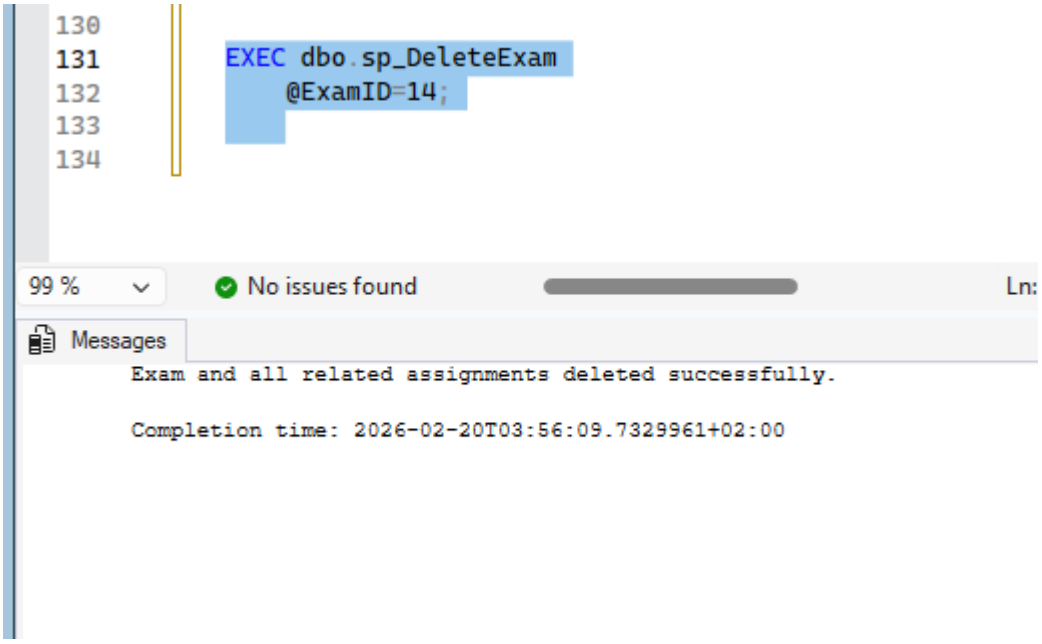
Deletes an exam and its related exam-question records safely.

🧾 Parameters

| @ExamID INT |

🧪 Test Query

```
EXEC dbo.sp_DeleteExam
    @ExamID=2;
```



✅ Expected Result

Exam and related ExamQuestion rows removed.

🔎 Applied To

- Exam
- ExamQuestion

9️⃣ dbo.sp_DeleteIntake_Hard

📄 Description

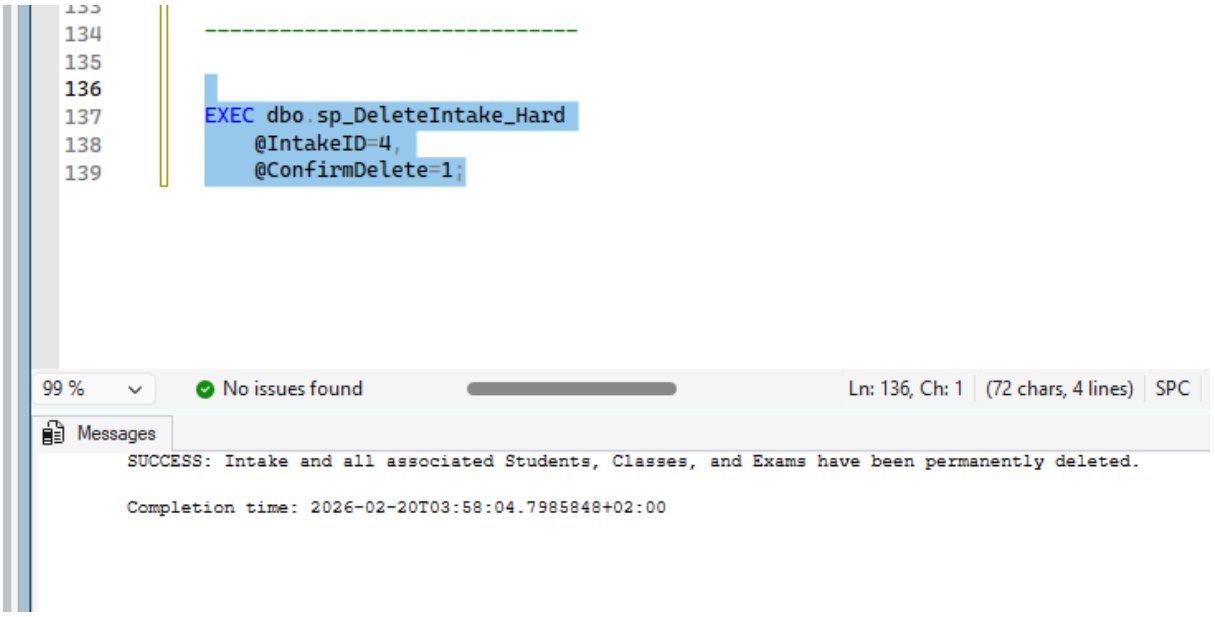
Deletes intake after confirmation flag validation.

🧾 Parameters

| @IntakeID INT | @ConfirmDelete BIT |

🧪 Test Query

```
EXEC dbo.sp_DeleteIntake_Hard
    @IntakeID=4,
    @ConfirmDelete=1;
```



🔎 Applied To

- Intake

🔟 dbo.sp_DeleteStudent_Hard

📄 Description

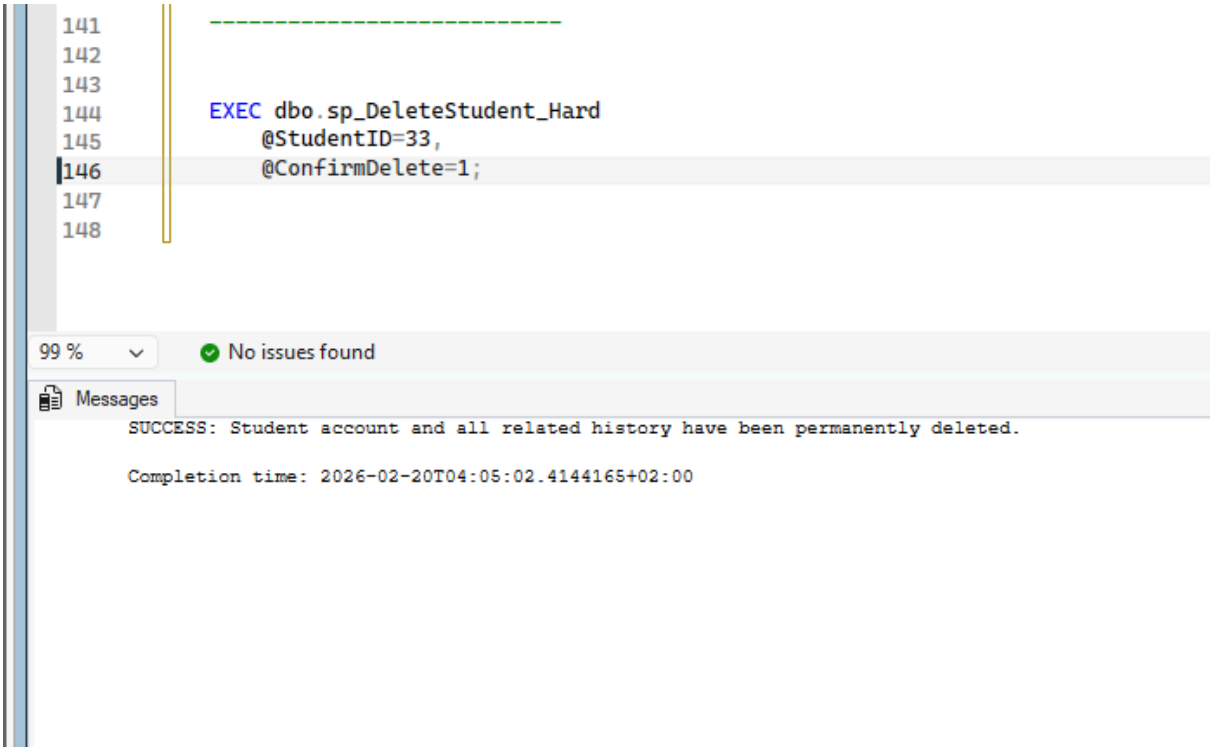
Permanently removes student and related enrollment data.

🧾 Parameters

| @StudentID INT | @ConfirmDelete BIT |

🧪 Test Query

```
EXEC dbo.sp_DeleteStudent_Hard
    @StudentID=33,
    @ConfirmDelete=1;
```



🔎 Applied To

- Student
- StudentExam

1️⃣1️⃣ dbo.sp_DeleteTrack_Hard

📄 Description

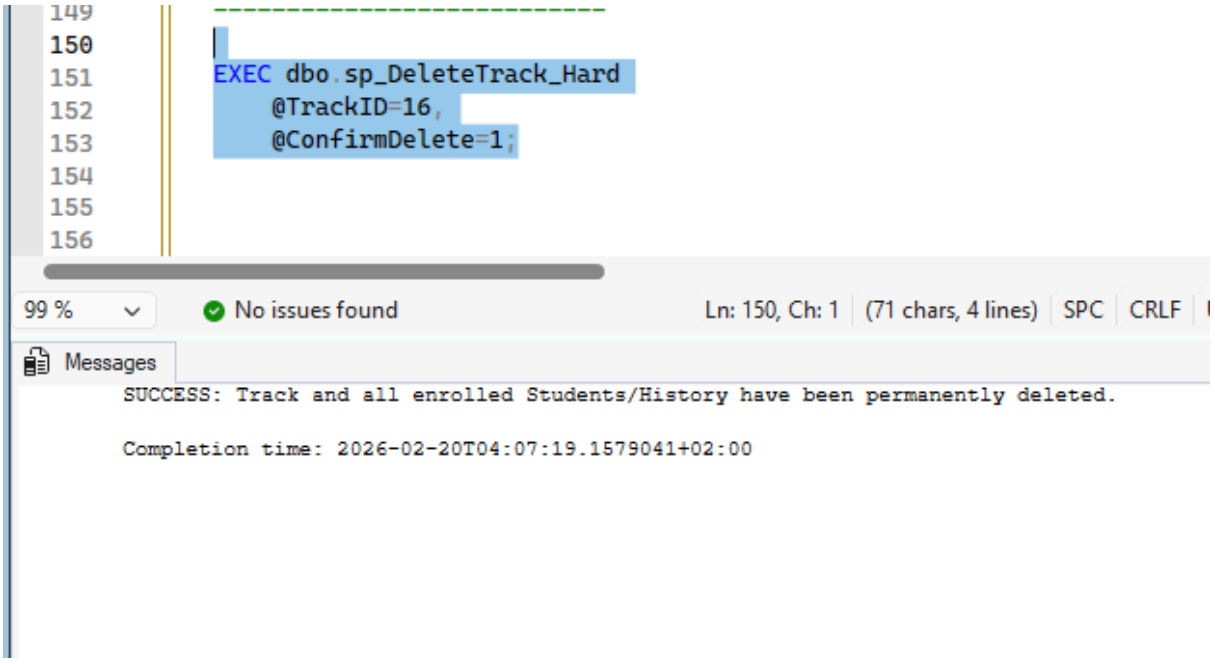
Deletes a training track after confirmation.

🧾 Parameters

| @TrackID INT | @ConfirmDelete BIT |

🧪 Test Query

```
EXEC dbo.sp_DeleteTrack_Hard
    @TrackID=2,
    @ConfirmDelete=1;
```



🔎 Applied To

- Track

1️⃣2️⃣ dbo.sp_EditBranch

📄 Description

Updates branch name.

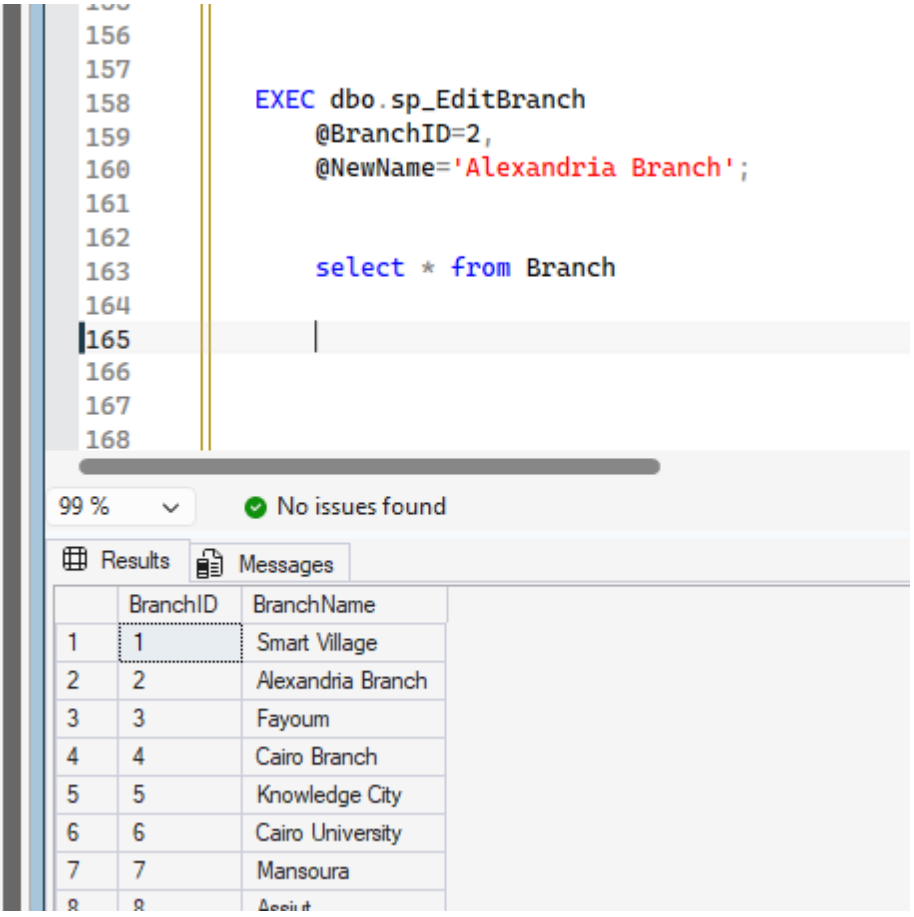
🧾 Parameters

| @BranchID INT | @NewName VARCHAR(25) |

🧪 Test Query

```
EXEC dbo.sp_EditBranch
    @BranchID=2,
    @NewName='Alexandria Branch';

    select * from Branch
```



🔎 Applied To

- Branch

1️⃣3️⃣ dbo.sp_EditIntake

📄 Description

Updates intake start and end dates.

🧾 Parameters

| @IntakeID INT | @NewStartDate DATE | @NewEndDate DATE |

🧪 Test Query

```
EXEC dbo.sp_EditIntake
    @IntakeID=2,
    @NewStartDate='2026-02-01',
    @NewEndDate='2026-07-01';
```

```
164
165
166        ----------------------------
167
168
169
170        EXEC dbo.sp_EditIntake
171        @IntakeID=7,
172        @NewStartDate='2026-02-01',
173        @NewEndDate='2026-07-01';
174
175        select * from Intake
```

99 % No issues found Ln: 175, Ch: 1 (24 chars, 1

Results | Messages

	IntakeID	StartDate	EnDate	ExamID	BranchID
1	2	2025-10-01	2026-06-30	2	1
2	3	2025-10-01	2026-06-30	3	2
3	5	2025-11-01	2026-08-01	NULL	4
4	6	2025-11-01	2026-08-01	NULL	7
5	7	2026-02-01	2026-07-01	NULL	8
6	8	2026-01-01	2026-06-01	NULL	1
7	9	2026-01-01	2026-06-01	NULL	1
8	10	2026-01-01	2026-06-01	NULL	1

🔎 Applied To

- Intake

1️⃣4️⃣ dbo.sp_EditQuestion

📄 Description

Updates question text, degree, and answer metadata.

🧾 Parameters

Parameter	Type
@QuestionID	INT

Parameter	Type
@NewQText	NVARCHAR(MAX)
@NewDegree	INT
@NewCorrectAnswer	NVARCHAR(MAX)
@NewWrongAnswers	NVARCHAR(MAX)
@NewIsTrue	BIT
@NewKeywords	NVARCHAR(MAX)

🧪 Test Query

```
EXEC dbo.sp_EditQuestion
    @QuestionID=5,
    @NewQText='Updated question?',
    @NewDegree=10,
    @NewCorrectAnswer='Correct',
    @NewWrongAnswers='Wrong1,Wrong2',
    @NewIsTrue=NULL,
    @NewKeywords=NULL;
```

```
178 ------------------------------------
179
180
181
182 EXEC dbo.sp_EditQuestion
183     @QuestionID=5,
184     @NewQText='Updated question?',
185     @NewDegree=10,
186     @NewCorrectAnswer='Correct',
187     @NewWrongAnswers='Wrong1,Wrong2',
188     @NewIsTrue=NULL,
189     @NewKeywords=NULL;
```

99 % | No issues found | Ln: 182, Ch: 1 | (219 chars,

Messages

```
Question and Answer updated successfully.

Completion time: 2026-02-20T04:40:14.2070778+02:00
```

🔎 Applied To

- Question
- MCQ / TF / Keywords tables

1️⃣5️⃣ Stored Procedure: dbo.sp_EditTrack

Type: Management Procedure

📄 Description

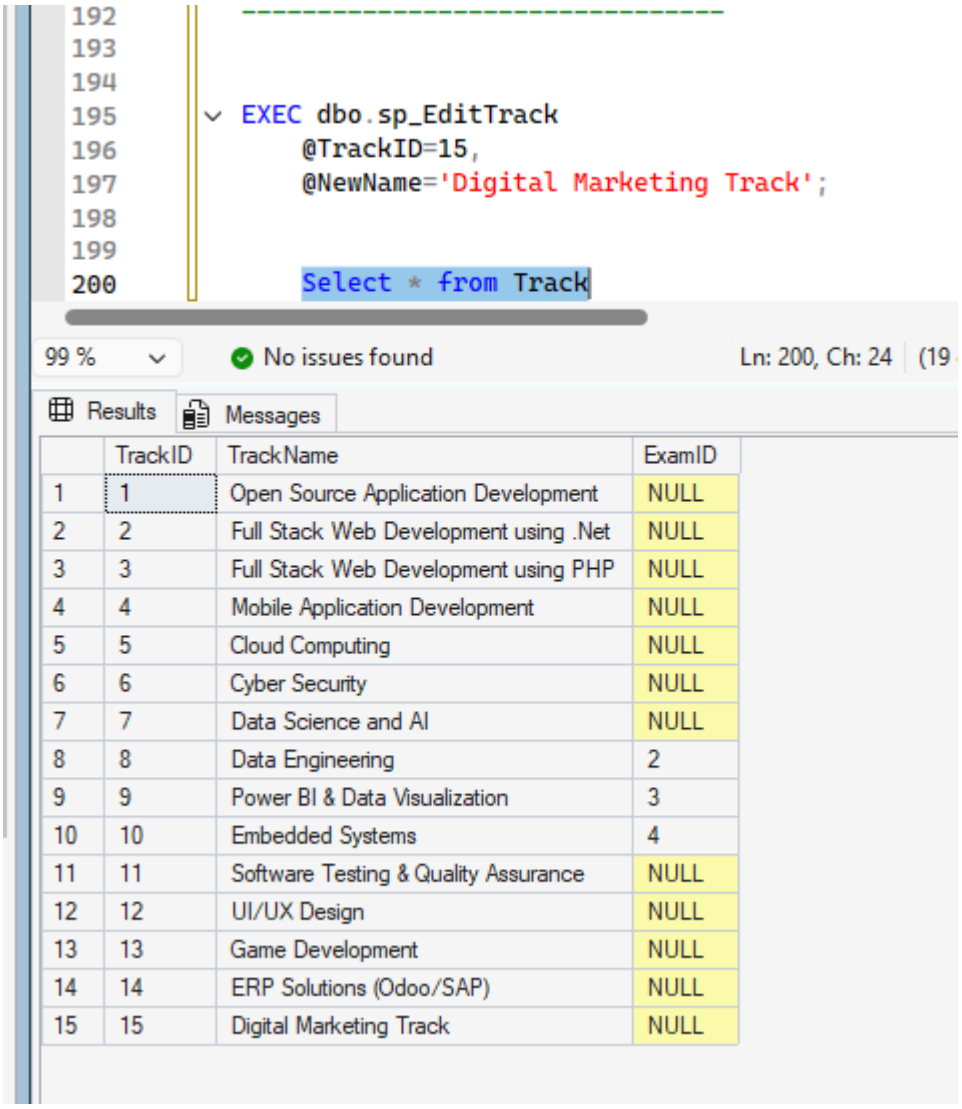
Updates the name of an existing training track.

🧾 Parameters

Parameter	Data Type
@TrackID	INT
@NewName	NVARCHAR(140)

🧪 Test Query

```
EXEC dbo.sp_EditTrack
    @TrackID=1,
    @NewName='Data Engineering';
```



	TrackID	TrackName	ExamID
1	1	Open Source Application Development	NULL
2	2	Full Stack Web Development using .Net	NULL
3	3	Full Stack Web Development using PHP	NULL
4	4	Mobile Application Development	NULL
5	5	Cloud Computing	NULL
6	6	Cyber Security	NULL
7	7	Data Science and AI	NULL
8	8	Data Engineering	2
9	9	Power BI & Data Visualization	3
10	10	Embedded Systems	4
11	11	Software Testing & Quality Assurance	NULL
12	12	UI/UX Design	NULL
13	13	Game Development	NULL
14	14	ERP Solutions (Odoo/SAP)	NULL
15	15	Digital Marketing Track	NULL

✅ Expected Result

Track name updated successfully.

🔎 Applied To Which Tables

- Track

1️⃣6️⃣ Stored Procedure: dbo.sp_FinalExamResult

Type: Business Logic Procedure

📄 Description

Calculates and returns the student's final result in a specific exam and may update course result.

🧾 Parameters

Parameter	Data Type
@StudentID	INT
@ExamID	INT

🧪 Test Query

```
EXEC dbo.sp_FinalExamResult
    @StudentID=7,
    @ExamID=2;
```

```sql
EXEC dbo.sp_FinalExamResult
    @StudentID=7,
    @ExamID=2;
```

99 % No issues found Ln: 209

Results | Messages

	StudentName	CourseName	ScoreObtained	MaxScore	Status
1	Omar Farouk	SQL Server Fundamentals	0	100	Failed

✅ Expected Result

Returns final calculated score and pass/fail status.

🔎 Applied To Which Tables

- StudentAnswer
- CourseResult
- Exam

1️⃣7️⃣Stored Procedure: dbo.sp_GetStudentResult

Type: Reporting Procedure

📄 Description

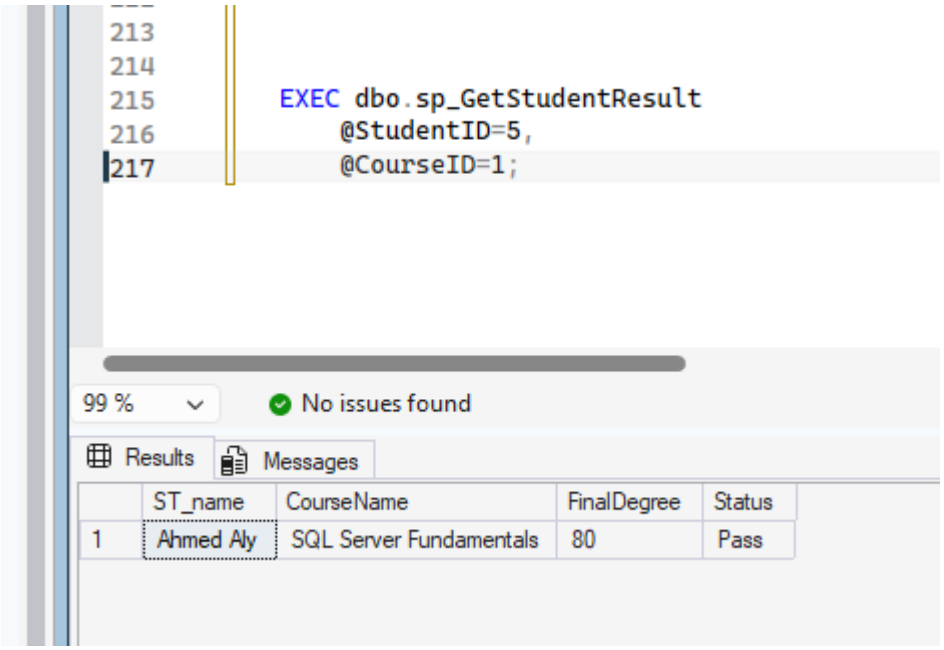
Returns a student's final result in a specific course.

🧾 Parameters

Parameter	Data Type
@StudentID	INT
@CourseID	INT

🧪 Test Query

```
EXEC dbo.sp_GetStudentResult
    @StudentID=10,
    @CourseID=1;
```



✅ Expected Result

Returns final course grade and status.

🔎 Applied To Which Tables

- CourseResult
- Course

1️⃣8️⃣ Stored Procedure: dbo.sp_ins_viewExam

Type: Instructor View Procedure

📄 Description

Returns exam details for instructor workflow.

🧾 Parameters

Parameter	Data Type
@InstructorID	INT
@ExamID	INT

🧪 Test Query

```
EXEC dbo.sp_ins_viewExam
    @InstructorID=2,
    @ExamID=5;
```

	ExamID	CourseName	Exam Date	StartTime	QuestionID	Question	Type	Marks	Choices	Model Answer
1	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	1	Which of the following is used to remove all rec...	MCQ	5	Truncate, Delete, Drop, Remove	Truncate
2	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	4	Which join returns all rows when there is a matc...	MCQ	5	Inner Join, Full Outer Join, Left Join, Cross Join	Full Outer Join
3	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	5	Updated question?	TF	10	True, False	False
4	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	9	Explain the concept of Normalization and its imp...	Text	10	N/A	Data Redundancy, Data Integrity, Anomalies

---

✅ Expected Result

Returns exam details only if instructor owns the exam.

🔎 Applied To Which Tables

- Exam
- Class
- Instructor

---

19 Stored Procedure: dbo.sp_RegisterStudent

Type: Registration Procedure

📄 Description

Creates a login account and student enrollment profile.

🧾 Parameters

Parameter	Data Type
@Username	VARCHAR(50)
@Password	VARCHAR(50)
@FullName	VARCHAR(50)
@Email	VARCHAR(75)
@BranchID	INT
@TrackID	INT
@IntakeID	INT

🧪 Test Query

```
EXEC dbo.sp_RegisterStudent
    @Username='mohamed1',
    @Password='123456',
    @FullName='Mohamed Ali',
    @Email='mohamed@gmail.com',
    @BranchID=1,
    @TrackID=1,
    @IntakeID=1;
```

```sql
EXEC dbo.sp_RegisterStudent
    @Username='mohamed1',
    @Password='123456',
    @FullName='Mohamed Ali',
    @Email='mohamed@gmail.com',
    @BranchID=2,
    @TrackID=8,
    @IntakeID=5;

    select * from [User] WHERE Username = 'mohamed1'
```

99 % No issues found Ln: 238, Ch: 4

Results | Messages

	UserID	Username	Role	Password
1	54	mohamed1	Student	123456

🔎 Applied To Which Tables

- User
- Student

---

21 Stored Procedure: dbo.sp_RemoveQuestion

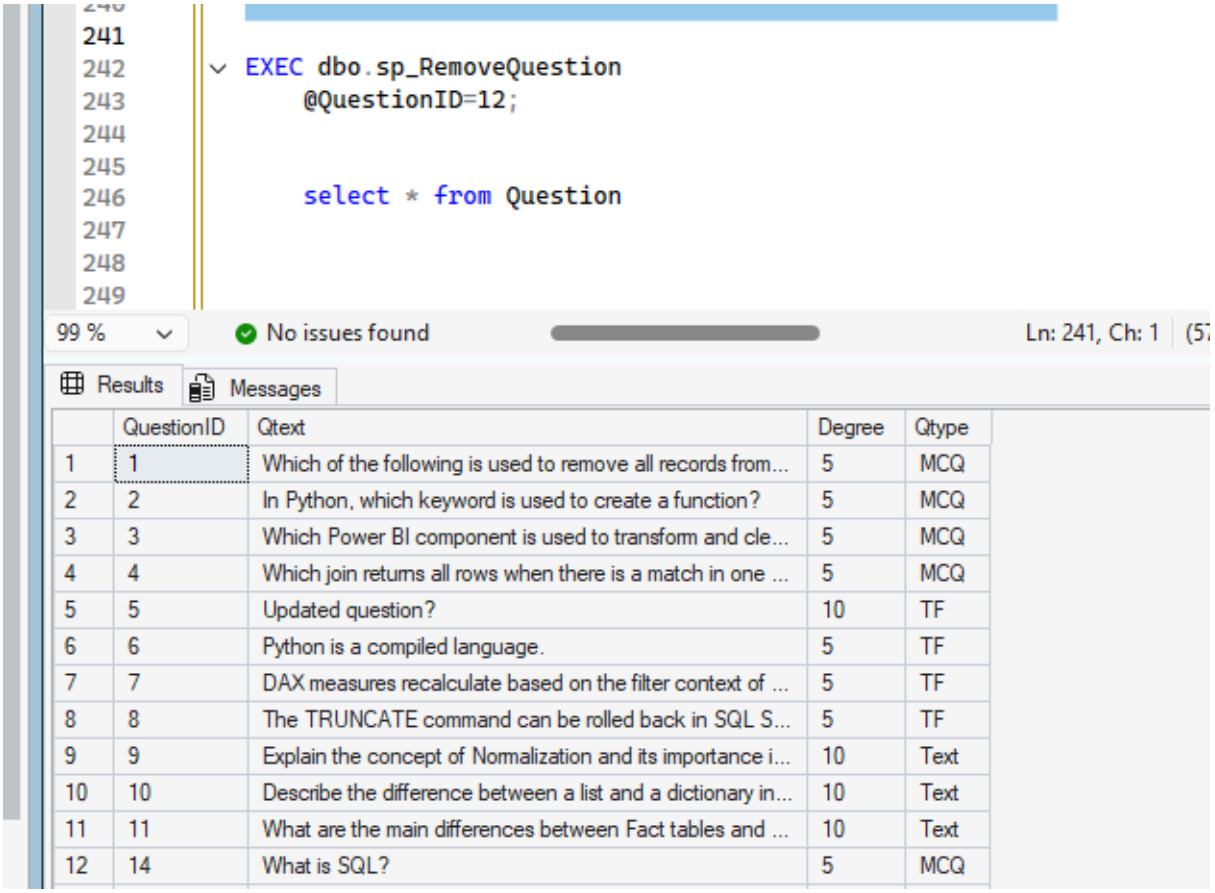
🧾 Parameters

Parameter	Data Type
@QuestionID	INT

🧪 Test Query

```sql
EXEC dbo.sp_RemoveQuestion
```

```
    @QuestionID=4;
```



	QuestionID	Qtext	Degree	Qtype
1	1	Which of the following is used to remove all records from...	5	MCQ
2	2	In Python, which keyword is used to create a function?	5	MCQ
3	3	Which Power BI component is used to transform and cle...	5	MCQ
4	4	Which join returns all rows when there is a match in one ...	5	MCQ
5	5	Updated question?	10	TF
6	6	Python is a compiled language.	5	TF
7	7	DAX measures recalculate based on the filter context of ...	5	TF
8	8	The TRUNCATE command can be rolled back in SQL S...	5	TF
9	9	Explain the concept of Normalization and its importance i...	10	Text
10	10	Describe the difference between a list and a dictionary in...	10	Text
11	11	What are the main differences between Fact tables and ...	10	Text
12	14	What is SQL?	5	MCQ

🔎 Applied To

- Question
- CourseQuestion

---

23 Stored Procedure: dbo.sp_SaveInstructor

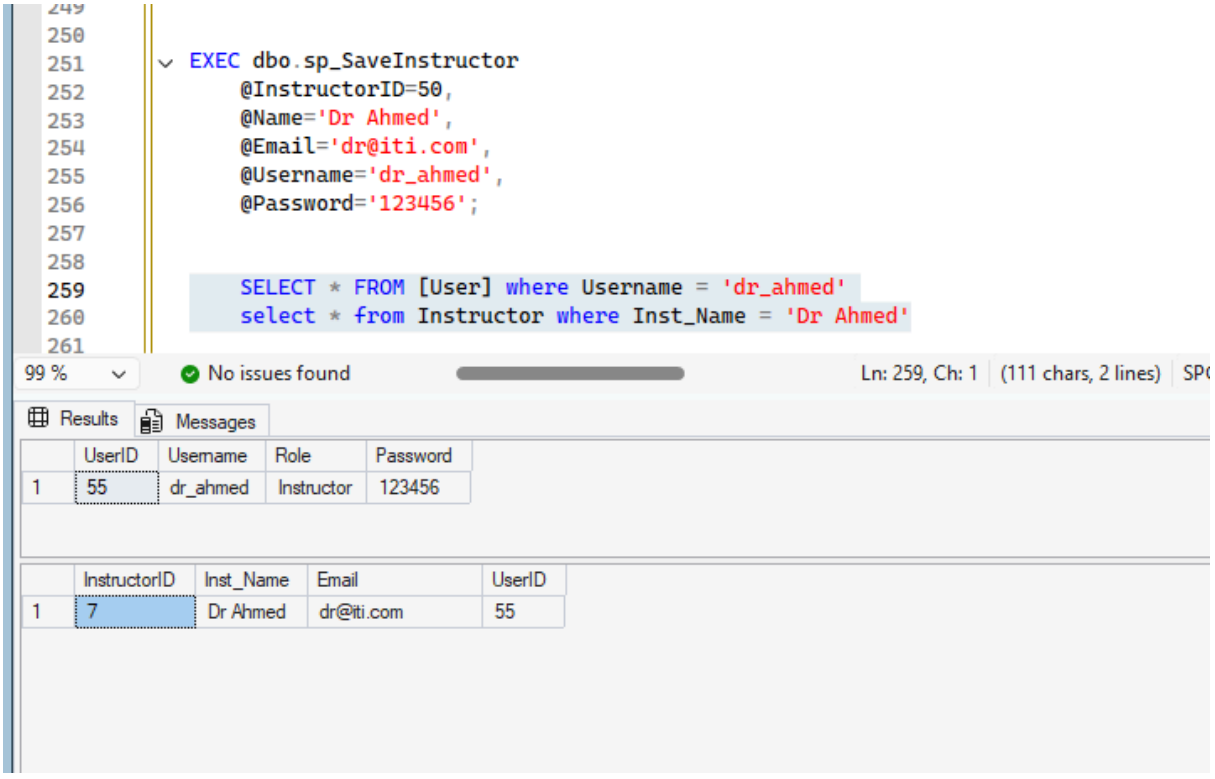
🧾 Parameters

Parameter	Data Type
@InstructorID	INT
@Name	NVARCHAR(50)
@Email	VARCHAR(75)

Parameter	Data Type
@Username	VARCHAR(50)
@Password	VARCHAR(50)

🧪 Test Query

```
EXEC dbo.sp_SaveInstructor
    @InstructorID=NULL,
    @Name='Dr Ahmed',
    @Email='dr@iti.com',
    @Username='dr_ahmed',
    @Password='123456';
```



🔎 Applied To

- Instructor
- User

---

24 StoredProcedure: dbo.sp_AssignStudentsToExam

Type: Adding Procedure

📄 Description

Assign specifc students in a track to a Exam (StudentID can be NULL)

🧾 Parameters

Parameter	Data Type
@ExamID	INT
@StudentID	INT
@TrackID	INT

🧪 Test Query

```
Exec sp_AssignStudentsToExam ReplaceThisWithNewExamID,Null,
8
```

```
Exec sp_AssignStudentsToExam 23,Null,8
```

108 % No issues found

Messages

```
4 students assigned to Exam 23.

Completion time: 2026-02-21T15:34:20.3651230+02:00
```

✅ Expected Result

Returns all number os the students assigned to this exam

---

25 Stored Procedure: dbo.sp_SumbitAnswer

Type: submit/adding answer procedure

📄 Description

Let the student sumbit the answers for his exam

🧾 Parameters

Parameter	Data Type
@ExamID	INT
@QuestionID	INT

Parameter	Data Type
@AnswerText	VARCHAR(max)

🧪 Test Query

```sql
USE ExamsSystem;
GO

DECLARE @StudentID INT = 25;
DECLARE @ExamID INT = 13;

-- ==========================================================================
-- PART 1: MULTIPLE CHOICE QUESTIONS
-- ==========================================================================

EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 4,  @AnswerText = 'Full Outer Join';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 13, @AnswerText = 'Structured Query Language';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 15, @AnswerText = 'UPDATE';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 17, @AnswerText = 'INNER JOIN';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 21, @AnswerText = 'COUNT()';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 23, @AnswerText = 'TRUNCATE';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 24, @AnswerText = 'ALTER';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 25, @AnswerText = 'HAVING';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 30, @AnswerText = 'SELECT DISTINCT';
EXEC sp_SubmitAnswer @StudentID, @ExamID, @QuestionID = 31, @AnswerText = 'master';
```

```sql
----------After stendtens submit answers we do grading--------------
Exec sp_FinalExamResult 25,13
-----
```

PROBLEMS 1 | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS | QUERY RESULTS

Results (1) | Messages

	StudentName	CourseName	ScoreObtained	MaxScore	Status
1	Hazem Emam	SQL Server Fundamentals	97	100	Passed

✅ Expected Result

Save the answers of the students in the database

🧩 SECTION 3 — VIEWS TESTING

1️⃣ View: dbo.v_adm_AllInstructor

📄 Description

Displays all instructor accounts for system administration monitoring.

📊 Columns

- UserID (int)
- Username (varchar)
- Role (varchar)
- Password (varchar)

🧪 Test Query

```
SELECT*FROM dbo.v_adm_AllInstructor;
```

Results | Messages

	UserID	Username	Role	Password
1	7	ahmed_instructor	Instructor	Inst_Pass_2026
2	9	inst_mona	Instructor	Mona@Pass2026
3	10	inst_yasser	Instructor	Yasser@Pass2026
4	11	inst_hoda	Instructor	Hoda@Pass2026
5	55	dr_ahmed	Instructor	123456

✅ Expected Result

Returns all users with role = Instructor.

🔎 Based On Tables

- User

2️⃣ dbo.v_adm_AllManager

📄 Description

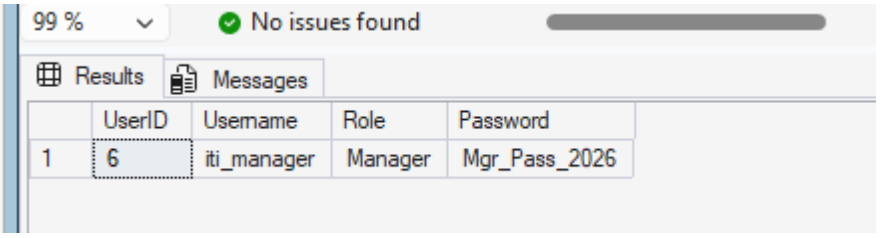
Shows all manager accounts for admin monitoring.

📊 Columns

UserID, Username, Role, Password

🧪 Test Query

```
SELECT*FROM dbo.v_adm_AllManager;
```



	UserID	Username	Role	Password
1	6	iti_manager	Manager	Mgr_Pass_2026

🔎 Based On

- User

3️⃣ dbo.v_adm_AllStudent

📄 Description

Displays all student login accounts.

📊 Columns

UserID, Username, Role, Password

```
SELECT*FROM dbo.v_adm_AllStudent;
```

```
278
279
280  SELECT*FROM dbo.v_adm_AllStudent;
```

99 % No issues found Ln: 280, C

Results | Messages

	UserID	Username	Role	Password
1	8	omar_student	Student	Std_Pass_2026
2	12	ahmed.aly	Student	kP9$vR2zQ!
3	13	mona.hassan	Student	bA4@mN7xC#
4	14	omar.farouk	Student	tD1&yU8sJ$
5	15	sara.mahmoud	Student	mG6*hK3vL@
6	16	yassin.ezzat	Student	wQ9#pB5zR!
7	17	nour.zaki	Student	xN2!fG8dW&
8	18	hany.salem	Student	rS5@vT1kP#
9	19	mariam.ali	Student	jK3$hL7mB!
10	20	khaled.idris	Student	vC6&pZ2sY@
11	21	laila.soliman	Student	fM8*rX4wQ#
12	22	hassan.reda	Student	sB1!nK9tD$
13	23	dina.moussa	Student	yW5#gH2vP&
14	24	mostafa.nabil	Student	zL7@jM4kS!
15	25	reem.adel	Student	hP3$dQ8xR#
16	26	tarek.abdelaziz	Student	uR6*yS1tN@
17	27	salma.bakr	Student	qT5#mN8vX!
18	28	fady.george	Student	zY2$bV9wK@
19	29	habiba.wael	Student	pL4&rD1sZ#
20	30	sherif.gamal	Student	nV3#kS6tP$
21	31	aya.ibrahim	Student	cJ8!mR4yQ&
22	32	amr.diab	Student	tB5@mW9zL#
23	33	mai.ezz	Student	hX1$vN6rK!

Based On: User

4️⃣ dbo.v_adm_AllUsers

📄 Description

Shows all system users in one unified list.

```
SELECT*FROM dbo.v_adm_AllUsers;
```

99 % No issues found Ln: 284, Ch: 1 (31 chars, 1 lines) SPC CRLF UTF-

Results Messages

	UserID	Username	Role	Password
1	2	Ahmed22	Admin	2026
2	5	itgate_admin	Admin	Admin@123
3	6	iti_manager	Manager	Mgr_Pass_2026
4	7	ahmed_instructor	Instructor	Inst_Pass_2026
5	8	omar_student	Student	Std_Pass_2026
6	9	inst_mona	Instructor	Mona@Pass2026
7	10	inst_yasser	Instructor	Yasser@Pass2026
8	11	inst_hoda	Instructor	Hoda@Pass2026
9	12	ahmed.aly	Student	kP9$vR2zQ!
10	13	mona.hassan	Student	bA4@mN7xC#
11	14	omar.farouk	Student	tD1&yU8sJ$
12	15	sara.mahmoud	Student	mG6*hK3vL@
13	16	yassin.ezzat	Student	wQ9#pB5zR!
14	17	nour.zaki	Student	xN2!fG8dW&
15	18	hany.salem	Student	rS5@vT1kP#
16	19	mariam.ali	Student	jK3$hL7mB!
17	20	khaled.idris	Student	vC6&pZ2sY@
18	21	laila.soliman	Student	fM8*rX4wQ#
19	22	hassan.reda	Student	sB1!nK9tD$
20	23	dina.moussa	Student	yW5#gH2vP&
21	24	mostafa.nabil	Student	zL7@jM4kS!
22	25	reem.adel	Student	hP3$dQ8xR#
23	26	tarek.abdelaziz	Student	uR6*yS1tN@

Based On: User

5️⃣ dbo.v_adm_DatabaseUsageStatistics

📄 Description

Provides summarized database counts by category.

📊 Columns

StatisticType, Category, TotalCount

```
SELECT*FROM dbo.v_adm_DatabaseUsageStatistics;
```

---

99 % No issues found Ln: 287, Ch: 1 (46 chars, 1 lines) SPC CRLF UTF

Results Messages

	StatisticType	Category	TotalCount
1	Exams Per Year	2026	3
2	Student Per Intake	5	1
3	Student Per Intake	2	28
4	Total Answers Stored	All Years	4

Based On: Student, Course, Exam, etc.

6️⃣ dbo.v_adm_SystemActivitySummary

📄 Description

Displays system-wide totals (students, instructors, exams, etc.).

📊 Columns

TotalStudents, TotalInstructors, TotalCourses, TotalExams, TotalDepartments, TotalTracks

```
SELECT*FROM dbo.v_adm_SystemActivitySummary;
```

```
290
291    SELECT*FROM dbo.v_adm_SystemActivitySummary;
```

99 % No issues found Ln: 291, Ch: 1

Results Messages

	TotalStudents	TotalInstructors	TotalCourses	TotalExams	TotalDepartments	TotalTracks
1	29	4	14	3	9	15

Based On: Multiple master tables

🔷 INSTRUCTOR VIEWS

7️⃣ dbo.v_ins_CourseDetails

📄 Description

Shows instructor course details with student and exam counts.

📊 Columns

CourseID, CourseName, MinDegree, IntakeID, Inst_Name, TotalStudents, TotalExams

```
SELECT*FROM dbo.v_ins_CourseDetails;
```

99 % No issues found Ln: 297, Ch: 1 (36 chars, 1 lines) SP

Results Messages

	CourseID	CourseName	MinDegree	IntakeID	Inst_Name	TotalStudents	TotalExams
1	1	SQL Server Fundamentals	50	2	NULL	15	1
2	2	Data Warehouse & ETL	50	NULL	NULL	0	0
3	3	Python for Data Science	50	2	NULL	13	1
4	4	Power BI Modeling	50	2	Mona Ezzat	0	1
5	5	Big Data Technologies	50	NULL	Yasser Mahmoud	0	0
6	6	Cloud Data Solutions	60	NULL	Hoda Mansour	0	0
7	7	Data Visualization	50	NULL	Dr Ahmed	0	0
8	8	Soft Skills	50	NULL	NULL	0	0
9	10	Test	50	NULL	NULL	0	0
10	12	SQL Advanced	60	NULL	NULL	0	0
11	13	Test	50	NULL	NULL	0	0
12	16	SQL Advanced	60	NULL	NULL	0	0
13	17	Test	50	NULL	NULL	0	0
14	20	SQL Advanced	60	NULL	NULL	0	0

Based On: Course, Class, Instructor, Student, Exam

8️⃣ dbo.v_ins_ExamDetails

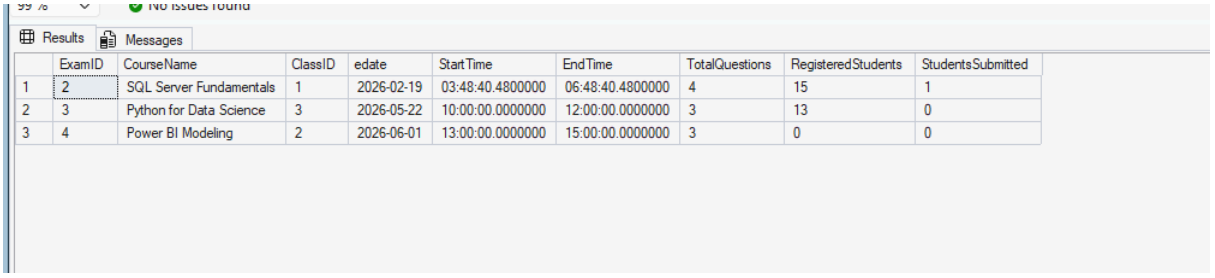
📄 Description

Displays exam participation and submission metrics.

📊 Columns

ExamID, CourseName, ClassID, edate, StartTime, EndTime, TotalQuestions, RegisteredStudents, StudentsSubmitted

```
SELECT*FROM dbo.v_ins_ExamDetails;
```



Results | Messages

	ExamID	CourseName	ClassID	edate	StartTime	EndTime	TotalQuestions	RegisteredStudents	StudentsSubmitted
1	2	SQL Server Fundamentals	1	2026-02-19	03:48:40.4800000	06:48:40.4800000	4	15	1
2	3	Python for Data Science	3	2026-05-22	10:00:00.0000000	12:00:00.0000000	3	13	0
3	4	Power BI Modeling	2	2026-06-01	13:00:00.0000000	15:00:00.0000000	3	0	0

Based On: Exam, ExamQuestion, StudentExam, StudentAnswer

9️⃣ dbo.v_Ins_InstructorCourses

📄 Description

Lists courses assigned to instructors.

📊 Columns

InstructorID, UserName, CourseName

```
SELECT*FROM dbo.v_Ins_InstructorCourses;
```

---

99 % No issues found

Results Messages

	InstructorID	UserName	CourseName
1	4	itgate_admin	SQL Server Fundamentals
2	5	iti_manager	Data Warehouse & ETL
3	6	ahmed_instructor	Python for Data Science

Based On: Instructor, Course, Inst_course

🔟 dbo.v_Ins_InstructorExamList

📄 Description

Shows exams created or owned by instructors.

📊 Columns

InstructorName, CourseName, ExamID, YearOfExam, AllowedDuration, ExamStatus

```
SELECT*FROM dbo.v_Ins_InstructorExamList;
```

---

99 % No issues found Ln: 308, Ch: 42 SPC CF

Results Messages

	InstructorName	CourseName	ExamID	YearOfExam	AllowedDuration	ExamStatus
1	Mona Ezzat	SQL Server Fundamentals	2	2026	Calculators allowed	Completed
2	Hoda Mansour	Python for Data Science	3	2026	Open book	Upcoming

---

Based On: Instructor, Exam, Class

1️⃣1️⃣ dbo.v_Ins_QuestionBankByInstructor

📄 Description

Displays question bank grouped by instructor and course.

📊 Columns

InstructorID, InstructorName, CourseID, CourseName, QuestionID, Qtext, Qtype, Degree

```
SELECT*FROM dbo.v_Ins_QuestionBankByInstructor;
```

```
311 ----------------------------------------
312
313 SELECT*FROM dbo.v_Ins_QuestionBankByInstructor;
```

99 % No issues found

Results | Messages

	InstructorID	InstructorName	CourseID	CourseName	QuestionID	Qtext	Qtype	Degree
1	4	itgate_admin	1	SQL Server Fundamentals	1	Which of the following is used to remove all recor...	MCQ	5
2	4	itgate_admin	1	SQL Server Fundamentals	4	Which join returns all rows when there is a match...	MCQ	5
3	4	itgate_admin	1	SQL Server Fundamentals	5	Updated question?	TF	10
4	4	itgate_admin	1	SQL Server Fundamentals	9	Explain the concept of Normalization and its impo...	Text	10
5	4	itgate_admin	1	SQL Server Fundamentals	14	What is SQL?	MCQ	5
6	4	itgate_admin	1	SQL Server Fundamentals	16	What is SQL?	MCQ	5
7	4	itgate_admin	1	SQL Server Fundamentals	18	What is SQL?	MCQ	5
8	6	ahmed_instructor	3	Python for Data Science	2	In Python, which keyword is used to create a fun...	MCQ	5
9	6	ahmed_instructor	3	Python for Data Science	6	Python is a compiled language.	TF	5
10	6	ahmed_instructor	3	Python for Data Science	10	Describe the difference between a list and a dicti...	Text	10

Based On: Instructor, Course, Question, CourseQuestion

1️⃣2️⃣ View: dbo.v_man_Branches

📄 Description

Provides a simplified list of all available branches for manager monitoring and reporting purposes.

📊 Columns

- BranchID (int)
- BranchName (varchar)

🧪 Test Query

```
SELECT*FROM dbo.v_man_Branches;
```

```
319 SELECT*FROM dbo.v_man_Branches;
```

99 % No issues found

Results | Messages

	BranchID	BranchName
1	1	Smart Village
2	2	Alexandria Branch
3	3	Fayoum
4	4	Cairo Branch
5	5	Knowledge City
6	6	Cairo University
7	7	Mansoura
8	8	Assiut
9	9	Menoufia
10	10	Ismailia
11	11	Minya
12	12	Aswan
13	13	Qena
14	14	Sohag
15	15	Tanta
16	16	Zagazig
17	17	Beni Suef
18	18	Port Said
19	19	Damietta
20	20	Banha
21	21	Damanhour
22	22	Al Arish

✅ Expected Result

Returns all branches in the system.

🔎 Based On Tables

- Branch

1️⃣3️⃣ View: dbo.v_man_IntakeDistribution

📄 Description

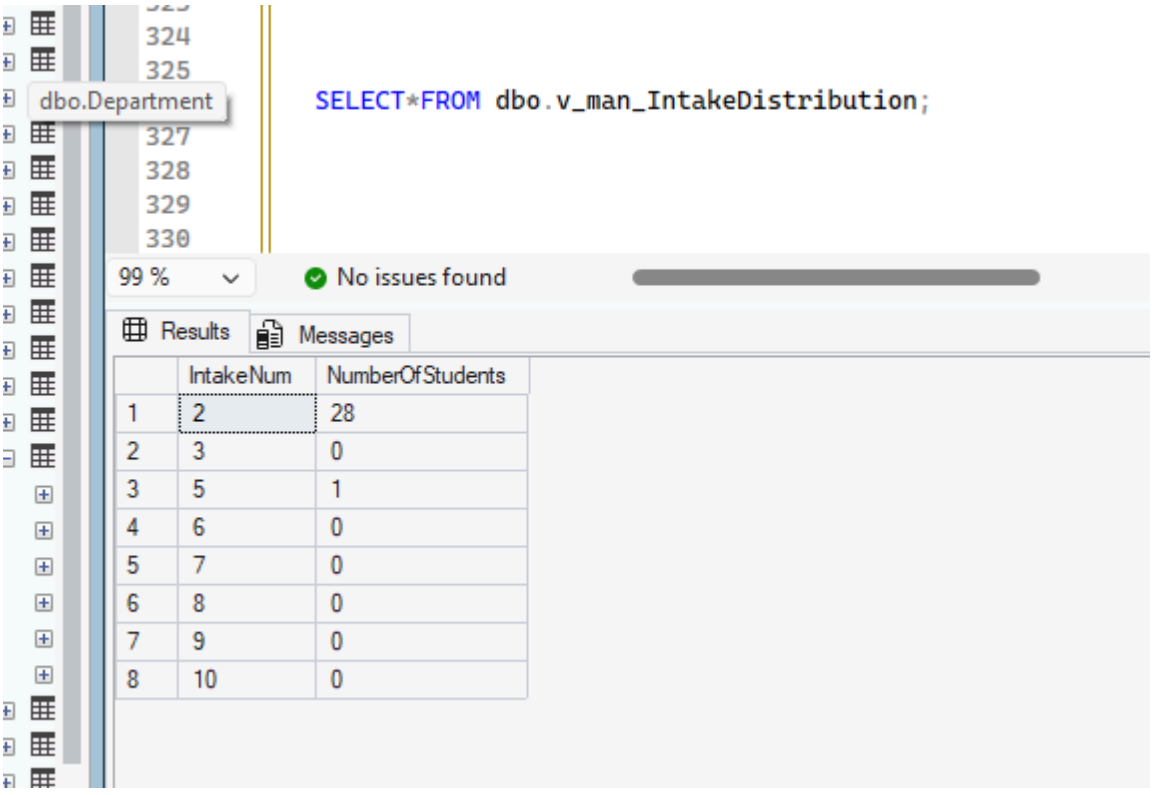
Displays the number of students enrolled in each intake period to help managers analyze enrollment distribution.

📊 Columns

- IntakeNum (int)
- NumberOfStudents (int)

🧪 Test Query

```
SELECT*FROM dbo.v_man_IntakeDistribution;
```



	IntakeNum	NumberOfStudents
1	2	28
2	3	0
3	5	1
4	6	0
5	7	0
6	8	0
7	9	0
8	10	0

✅ Expected Result

Returns each intake with total number of students enrolled.

🔎 Based On Tables

- Intake
- Student

1️⃣4️⃣ View: dbo.v_man_Intakes

📄 Description

Lists all intake records including their associated branch and exam information.

📊 Columns

- IntakeID (int)
- StartDate (date)
- EnDate (date)
- ExamID (int)
- BranchID (int)

🧪 Test Query

```
SELECT*FROM dbo.v_man_Intakes;
```

```
-----------------------------

SELECT*FROM dbo.v_man_Intakes;
```

99 % No issues found

Results | Messages

	IntakeID	StartDate	EnDate	ExamID	BranchID
1	2	2025-10-01	2026-06-30	2	1
2	3	2025-10-01	2026-06-30	3	2
3	5	2025-11-01	2026-08-01	NULL	4
4	6	2025-11-01	2026-08-01	NULL	7
5	7	2026-02-01	2026-07-01	NULL	8
6	8	2026-01-01	2026-06-01	NULL	1
7	9	2026-01-01	2026-06-01	NULL	1
8	10	2026-01-01	2026-06-01	NULL	1

🔎 Based On Tables

- Intake

1️⃣5️⃣ View: dbo.v_man_StudentPerformanceRanking

📄 Description

Ranks students based on their aggregated total scores across courses, allowing managers to evaluate performance standings.

📊 Columns

- StudentID (int)
- Username (varchar)
- TotalScore (int)
- RankPosition (bigint)

🧪 Test Query

```
SELECT*FROM dbo.v_man_StudentPerformanceRanking;
```

---

99 % No issues found

Results | Messages

	StudentID	Username	TotalScore	RankPosition
1	5	ahmed.aly	20	1

🔎 Based On Tables

- Student
- User
- CourseResult

1️⃣6️⃣ View: dbo.v_man_StudentsPerTrackBranch

📄 Description

Shows the number of students grouped by branch and training track for managerial analysis.

📊 Columns

- BranchName (varchar)
- TrackName (nvarchar)
- TotalStudents (int)

🧪 Test Query

```
SELECT*FROM dbo.v_man_StudentsPerTrackBranch;
```

99 % No issues found Ln: 342,

Results Messages

	BranchName	TrackName	TotalStudents
1	Alexandria Branch	Data Engineering	15
2	Alexandria Branch	Power BI & Data Visualization	14
3	Cairo Branch	Data Engineering	15
4	Cairo Branch	Power BI & Data Visualization	14
5	Fayoum	Data Engineering	15
6	Fayoum	Power BI & Data Visualization	14
7	Smart Village	Data Engineering	15
8	Smart Village	Power BI & Data Visualization	14

🔎 Based On Tables

- Student
- Branch
- Track

1️⃣7️⃣ View: dbo.v_man_TrackPerformanceSummary

📄 Description

Summarizes performance statistics (average, maximum, minimum scores) per track and branch.

📊 Columns

- TrackName (nvarchar)
- BranchName (varchar)

- AverageTotalScore (int)
- MaxTotalScore (int)
- MinTotalScore (int)

🧪 Test Query

```
SELECT*FROM dbo.v_man_TrackPerformanceSummary;
```

---

99 % No issues found

Results | Messages

	TrackName	BranchName	AverageTotalScore	MaxTotalScore	MinTotalScore
1	Data Engineering	Smart Village	20	20	20

🔎 Based On Tables

- Student
- CourseResult
- Track
- Branch

1️⃣8️⃣ View: dbo.v_man_TracksPerDepartment

📄 Description

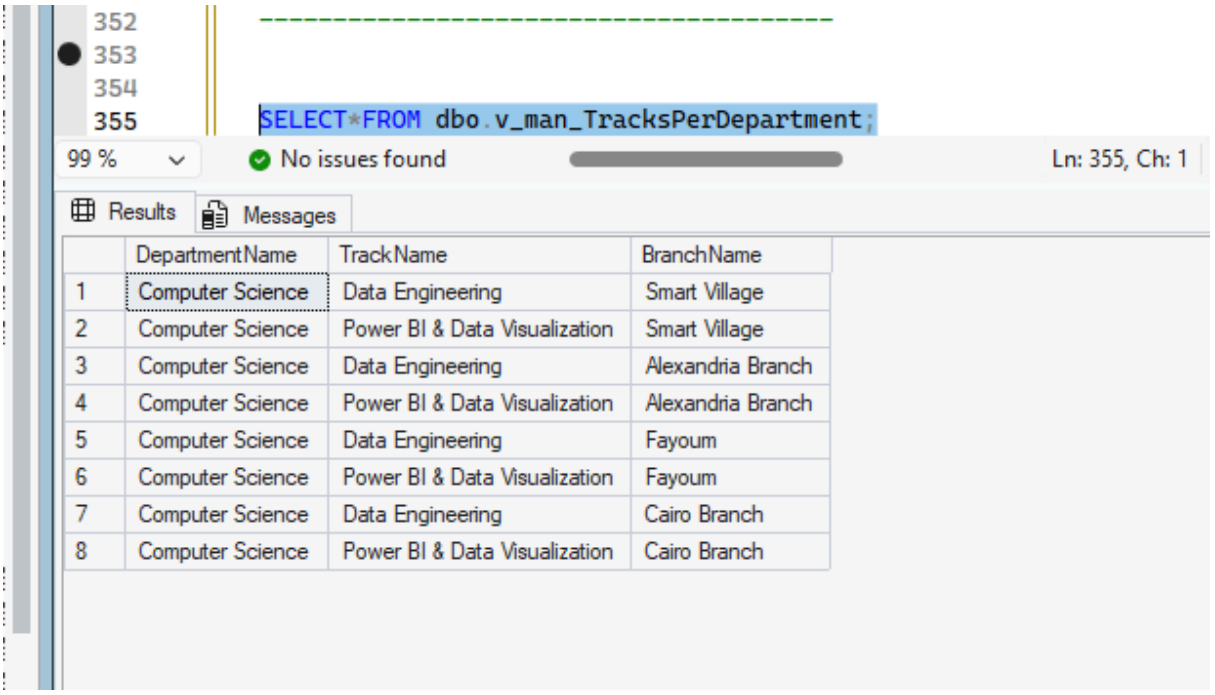
Displays the mapping between departments, tracks, and branches to clarify organizational structure.

📊 Columns

- DepartmentName (varchar)
- TrackName (nvarchar)
- BranchName (varchar)

🧪 Test Query

```
SELECT*FROM dbo.v_man_TracksPerDepartment;
```



	DepartmentName	TrackName	BranchName
1	Computer Science	Data Engineering	Smart Village
2	Computer Science	Power BI & Data Visualization	Smart Village
3	Computer Science	Data Engineering	Alexandria Branch
4	Computer Science	Power BI & Data Visualization	Alexandria Branch
5	Computer Science	Data Engineering	Fayoum
6	Computer Science	Power BI & Data Visualization	Fayoum
7	Computer Science	Data Engineering	Cairo Branch
8	Computer Science	Power BI & Data Visualization	Cairo Branch

---

🔎 Based On Tables

- Department
- Track
- TrackDepartment
- Branch

🔷 STUDENT VIEWS

1️⃣9️⃣ View: dbo.v_stu_AvailableExams

📄 Description

Shows exams available to each student including date, time, status, and submission state.

📊 Columns

- StudentID (int)
- ExamID (int)
- CourseName (varchar)
- edate (date)
- StartTime (time)
- EndTime (time)
- ExamStatus (varchar)
- SubmissionStatus (varchar)

🧪 Test Query

```
SELECT*FROM dbo.v_stu_AvailableExams;
```

```
SELECT*FROM dbo.v_stu_AvailableExams;
```

99 % No issues found

Results | Messages

	StudentID	ExamID	CourseName	edate	StartTime	EndTime	ExamStatus	SubmissionStatus
1	5	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Submitted
2	6	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
3	7	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
4	8	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
5	9	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
6	10	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
7	11	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
8	12	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
9	13	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
10	14	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
11	15	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
12	16	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
13	17	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
14	18	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
15	19	2	SQL Server Fundamentals	2026-02-19	03:48:40.4800000	06:48:40.4800000	Closed	Not Submitted
16	20	3	Python for Data Science	2026-05-22	10:00:00.0000000	12:00:00.0000000	Upcoming	Not Submitted
17	21	3	Python for Data Science	2026-05-22	10:00:00.0000000	12:00:00.0000000	Upcoming	Not Submitted
18	22	3	Python for Data Science	2026-05-22	10:00:00.0000000	12:00:00.0000000	Upcoming	Not Submitted
19	23	3	Python for Data Science	2026-05-22	10:00:00.0000000	12:00:00.0000000	Upcoming	Not Submitted

---

🔎 Based On Tables

- StudentExam
- Exam
- Course
- StudentAnswer

2️⃣0️⃣ View: dbo.v_stu_StudentAnswerReview

📄 Description

Allows students to review their submitted answers and compare them with the model answers after exam completion.

📊 Columns

- STID (int)
- ExamID (int)
- QText (text)
- IsCorrect (bit)
- ModelAnswer (nvarchar)
- StudentAnswer (nvarchar)

🧪 Test Query

```
SELECT*FROM dbo.v_stu_StudentAnswerReview;
```

```
364
365 SELECT*FROM dbo.v_stu_StudentAnswerReview;
```

99 % No issues found Ln: 365, Ch: 18 SPC CRLF UTF-

Results Messages

	STID	ExamID	QText	IsCorrect	ModelAnswer	StudentAnswer
1	5	2	Which of the following is used to remove all rec...	1	Truncate	TRUNCATE
2	5	2	Which join returns all rows when there is a matc...	1	Full Outer Join	Full Join
3	5	2	Updated question?	0	Review Required / Subjective	True
4	5	2	Updated question?	0	Review Required / Subjective	True
5	5	2	Explain the concept of Normalization and its imp...	1	Review Required / Subjective	Normalization reduces redundancy...

🔎 Based On Tables

- StudentAnswer
- Question
- MCQ / TrueFalseQuestion / TextQuestionKeywords

2️⃣1️⃣ View: dbo.v_Stu_StudentExamResults

📄 Description

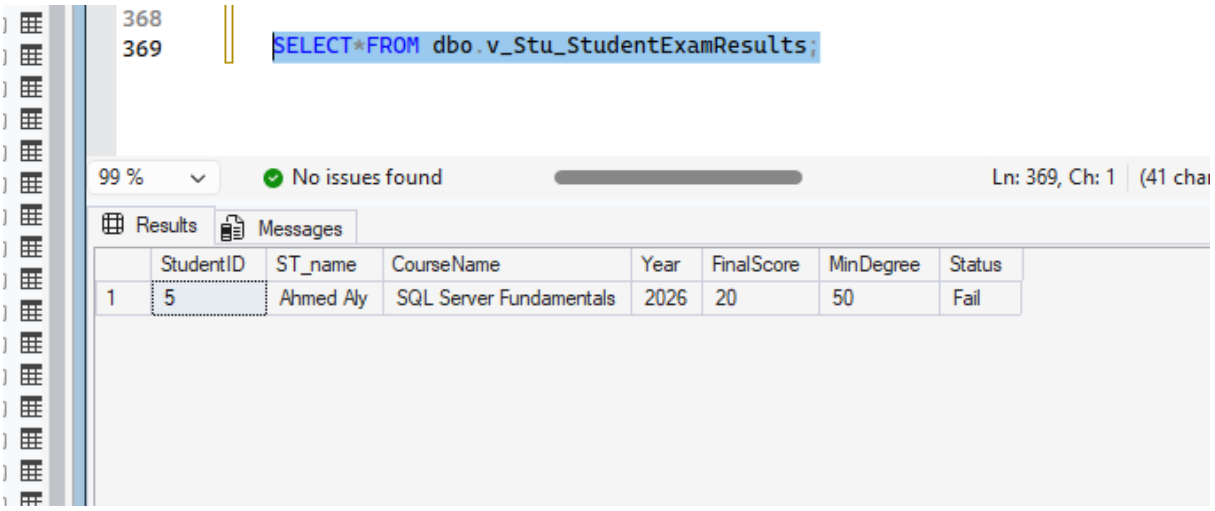
Displays final exam and course results for each student including score and pass/fail status.

📊 Columns

- StudentID (int)
- ST_name (varchar)
- CourseName (varchar)
- Year (int)
- FinalScore (int)
- MinDegree (int)
- Status (varchar)

🧪 Test Query

```
SELECT*FROM dbo.v_Stu_StudentExamResults;
```



🔎 Based On Tables

- Student
- CourseResult
- Course
- Class

🔷 GENERAL REPORTING VIEWS

2️⃣2️⃣ View: dbo.View_ExamQuestions

📄 Description

Displays exam questions along with their marks and answer choices for reporting and printing purposes.

📊 Columns

- ExamID (int)
- CourseName (varchar)

- QuestionID (int)
- Question (text)
- Type (varchar)
- Marks (int)
- Choices (nvarchar)

🧪 Test Query

```
SELECT*FROM dbo.View_ExamQuestions;
```

```
371
372 SELECT*FROM dbo.View_ExamQuestions;
```

99 % ✔ No issues found

Results | Messages

	ExamID	CourseName	QuestionID	Question	Type	Marks	Choices
1	2	SQL Server Fundamentals	1	Which of the following is used to remove all record...	MCQ	5	Truncate, Delete, Drop, Remove
2	2	SQL Server Fundamentals	4	Which join returns all rows when there is a match i...	MCQ	5	Inner Join, Full Outer Join, Left Join, Cross Join
3	2	SQL Server Fundamentals	5	Updated question?	TF	10	True, False
4	2	SQL Server Fundamentals	9	Explain the concept of Normalization and its import...	Text	10	Written Answer
5	3	Python for Data Science	2	In Python, which keyword is used to create a funct...	MCQ	5	func, define, def, function
6	3	Python for Data Science	6	Python is a compiled language.	TF	5	True, False
7	3	Python for Data Science	10	Describe the difference between a list and a dictio...	Text	10	Written Answer
8	4	Power BI Modeling	3	Which Power BI component is used to transform a...	MCQ	5	Power View, Power Pivot, Power Query, Power Serv...
9	4	Power BI Modeling	7	DAX measures recalculate based on the filter cont...	TF	5	True, False
10	4	Power BI Modeling	11	What are the main differences between Fact table...	Text	10	Written Answer

---

🔎 Based On Tables

- Exam
- ExamQuestion
- Question
- MCQ

2️⃣3️⃣ View: dbo.View_QuestionsWithModelAnswer

📄 Description

Displays question bank entries including their model answers for review and validation.

📊 Columns

- CourseName (varchar)
- Question (text)
- Type (varchar)
- Degree (int)
- Model Answer (nvarchar)

🧪 Test Query

```
SELECT*FROM dbo.View_QuestionsWithModelAnswers;
```

```
371
372
373  SELECT*FROM dbo.View_QuestionsWithModelAnswers;
```

99 % No issues found

Results | Messages

	CourseName	Question	Type	Degree	Model Answer
1	SQL Server Fundamentals	Which of the following is used to remove all record...	MCQ	5	Truncate
2	Python for Data Science	In Python, which keyword is used to create a funct...	MCQ	5	def
3	Power BI Modeling	Which Power BI component is used to transform a...	MCQ	5	Power Query
4	SQL Server Fundamentals	Which join returns all rows when there is a match i...	MCQ	5	Full Outer Join
5	SQL Server Fundamentals	Updated question?	TF	10	0
6	SQL Server Fundamentals	Updated question?	TF	10	0
7	Python for Data Science	Python is a compiled language.	TF	5	0
8	Power BI Modeling	DAX measures recalculate based on the filter cont...	TF	5	1
9	SQL Server Fundamentals	Explain the concept of Normalization and its import...	Text	10	Data Redundancy, Data Integrity, Anomalies
10	Python for Data Science	Describe the difference between a list and a dictio...	Text	10	Index-based access, Key-Value pairs, Mapping, Mu...
11	Power BI Modeling	What are the main differences between Fact table...	Text	10	Metrics, Foreign Keys, Star Schema, Slicing and Di...
12	SQL Server Fundamentals	What is SQL?	MCQ	5	Structured Query Language
13	SQL Server Fundamentals	What is SQL?	MCQ	5	Structured Query Language
14	SQL Server Fundamentals	What is SQL?	MCQ	5	Structured Query Language

---

🔎 Based On Tables

- Question
- Course

- MCQ
- TrueFalseQuestion

2️⃣5️⃣ View: dbo.View_StudentGrades

📄 Description

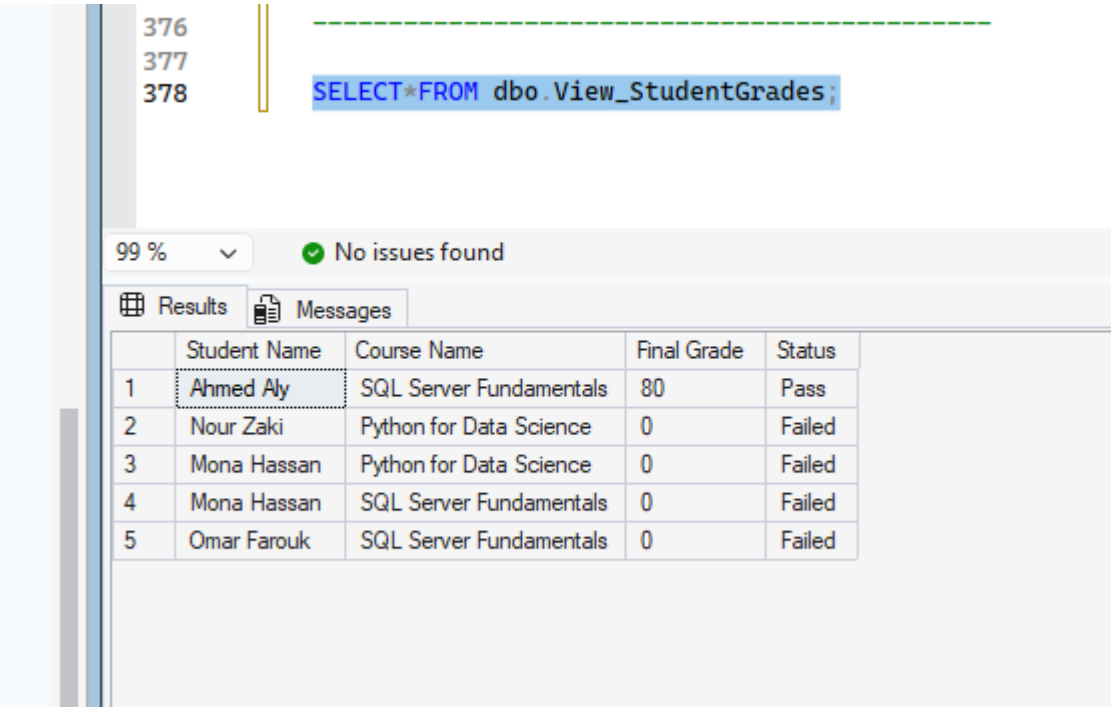
Shows final grades and pass/fail status for students per course.

📊 Columns

- Student Name (varchar)
- Course Name (varchar)
- Final Grade (int)
- Status (varchar)

🧪 Test Query

```
SELECT*FROM dbo.View_StudentGrades;
```



	Student Name	Course Name	Final Grade	Status
1	Ahmed Aly	SQL Server Fundamentals	80	Pass
2	Nour Zaki	Python for Data Science	0	Failed
3	Mona Hassan	Python for Data Science	0	Failed
4	Mona Hassan	SQL Server Fundamentals	0	Failed
5	Omar Farouk	SQL Server Fundamentals	0	Failed

🔎 Based On Tables

- Student
- CourseResult
- Course

---

26 View: dbo.vw_CourseClasses

📄 Description

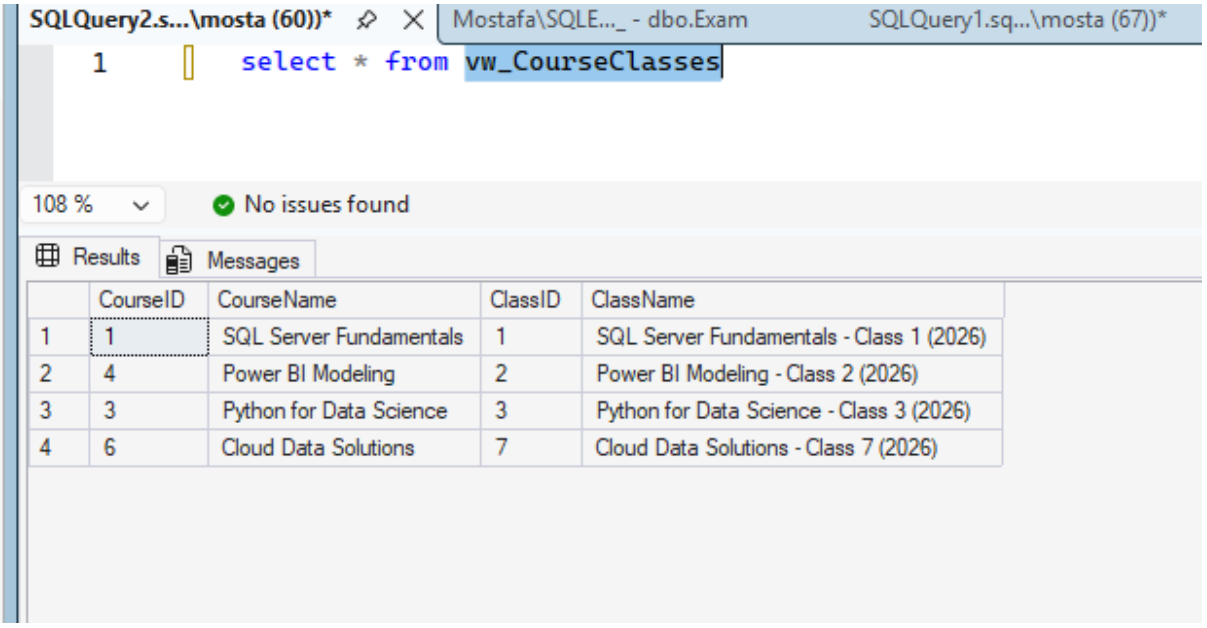
Shows the course Name and the class that is taking that course

📊 Columns

- Student Name (varchar)
- Course Name (varchar)
- Final Grade (int)
- Status (varchar)

🧪 Test Query

```
select * from vw_CourseClasses
```



🔎 Based On Tables

- CourseID
- CourseName
- ClassID
- ClassName

🔔 Database Triggers Documentation

1️⃣ trg_CheckExamTime_Submit

📌 Blocks answer submission outside exam time

📍 Table: `StudentAnswer`

🧪 Test Case (Invalid – After Exam End)

```
-- Try submitting answer after exam end time
INSERT INTO StudentAnswer ( isCorrect, Degree_given, Answer
text, QID, ExamID, STID)
VALUES (1,5,'Test Answer',1,2,10);
```

```sql
-- ## 1`dbo.trg_CheckExamTime_Submit`

INSERT INTO StudentAnswer ( isCorrect, Degree_given, Answertext, QID, ExamID, STID)
VALUES (1,5,'Test Answer',1,2,10);
```

No issues found

sages

```
Msg 50000, Level 16, State 1, Procedure trg_CheckExamTime_Submit, Line 26 [Batch Start Line 385]
Error: You cannot submit answers outside the scheduled exam date and time.
Msg 3609, Level 16, State 1, Line 386
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-02-20T14:17:57.2910015+02:00
```

✅ Expected Result

❌ Operation blocked

Error message like:

> Cannot submit answer outside exam time window.

2️⃣ trg_CheckExamWithinIntake

📌 Prevents exam date outside intake period

📍 Table: `Exam`

🧪 Test Case

```sql
-- Insert exam with date outside intake period
INSERT INTO Exam ( Allowence, ExamType, StartTime, EndTime,
ClassID, edate)
VALUES ('2 Hours','Normal','10:00','12:00',1,'2030-01-01');
```

```
389
390     ----- trg_CheckExamWithinIntake
391     -- Insert exam with date outside intake period
392     INSERT INTO Exam ( Allowence, ExamType, StartTime, EndTime, ClassID, edate)
393     VALUES ('2 Hours','Normal','10:00','12:00',1,'2030-01-01');
394
395
```

% No issues found

Messages

```
Msg 50000, Level 16, State 1, Procedure trg_CheckExamWithinIntake, Line 15 [Batch Start Line 391]
Error: The Exam Date must fall within the Intake's Start and End dates.
Msg 3609, Level 16, State 1, Line 392
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-02-20T14:22:16.2772161+02:00
```

✅ Expected Result

❌ Insert blocked

> Exam date must be within intake period.

3️⃣ trg_CheckGradeLimit

📌 Prevents negative or excessive grade

📍 Table: `StudentAnswer`

🧪 Test Case

```
-- Insert grade higher than question degree
INSERT INTO StudentAnswer (isCorrect, Degree_given, Answert
ext, QID, ExamID, STID)
VALUES (1,500,'Wrong Grade Test',1,2,10);
```

```
389
390    ----- trg_CheckExamWithinIntake
391    -- Insert exam with date outside intake period
392    INSERT INTO Exam ( Allowence, ExamType, StartTime, EndTime, ClassID, edate)
393    VALUES ('2 Hours','Normal','10:00','12:00',1,'2030-01-01');
394
395
```

% No issues found

Messages

```
Msg 50000, Level 16, State 1, Procedure trg_CheckExamWithinIntake, Line 15 [Batch Start Line 391]
Error: The Exam Date must fall within the Intake's Start and End dates.
Msg 3609, Level 16, State 1, Line 392
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-02-20T14:22:16.2772161+02:00
```

✅ Expected Result

❌ Operation blocked

> Grade exceeds allowed limit.

4️⃣ trg_CheckStudentRegistered

📌 Prevents unregistered student from answering

📍 Table: `StudentAnswer`

🧪 Test Case

```
-- Student not registered in StudentExam tries to answer

INSERT INTO StudentAnswer ( isCorrect, Degree_given, Answer
text, QID, ExamID, STID)
VALUES (1,5,'Test Answer',1,2,30);
```

```
402
403   -- trg_CheckStudentRegistered
404
405   INSERT INTO StudentAnswer ( isCorrect, Degree_given, Answertext, QID, ExamID, STID)
406   VALUES (1,5,'Test Answer',1,2,30);
407
```

99 % No issues found

Messages

```
Msg 50000, Level 16, State 1, Procedure trg_CheckExamTime_Submit, Line 26 [Batch Start Line 403]
Error: You cannot submit answers outside the scheduled exam date and time.
Msg 3609, Level 16, State 1, Line 405
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-02-20T14:28:13.4000947+02:00
```

✅ Expected Result

❌ Insert blocked

> Student is not registered for this exam.

---

5 trg_PreventDuplicateInstructorPerClass

📌 Prevent duplicate class definition

📍 Table: `Class`

🧪 Test Case

```
-- Insert duplicate class (same branch, intake, year, cours
e)
 INSERT INTO Class (Year, CourseID, Inst_ID, BranchID, Inta
keID)
VALUES (2026,1,5,1,2);
```

```
408
409
410
411  INSERT INTO Class (Year, CourseID, Inst_ID, BranchID, IntakeID)
412 VALUES (2026,1,5,1,2);
413
414
```

99 % No issues found

Messages

```
Msg 50000, Level 16, State 1, Procedure trg_PreventDuplicateInstructorPerClass, Line 22 [Batch Start Line 409]
Error: This course is already assigned in this class (Branch/Intake/Year). Duplicates are not allowed.
Msg 3609, Level 16, State 1, Line 411
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-02-20T14:37:02.9994747+02:00
```

✅ Expected Result

❌ Insert blocked

> Duplicate class definition not allowed.

---

6 trg_PreventExamOverlap

📌 Prevent overlapping exams

📍 Table: `Exam`

🧪 Test Case

```
-- Insert exam overlapping with existing one
INSERTINTO Exam (ExamID, Allowence, ExamType, StartTime, En
dTime, ClassID, edate)
VALUES (1000,'2 Hours','Mid','10:30','12:30',1,'2026-06-0
1');
```

```sql
--- trg_PreventExamOverlap

INSERT INTO Exam ( Allowence, ExamType, StartTime, EndTime, ClassID, edate)
VALUES ('Calculators allowed','Normal','03:48:40.4800000','06:48:40.4800000',1,'2026-02-19');
```

99 % No issues found

Messages

```
Msg 50000, Level 16, State 1, Procedure trg_PreventExamOverlap, Line 19 [Batch Start Line 417]
Exam time overlaps with another exam for the same class.
Msg 3609, Level 16, State 1, Line 418
The transaction ended in the trigger. The batch has been aborted.

Completion time: 2026-02-20T14:40:41.4008359+02:00
```

✅ Expected Result

❌ Insert blocked

> Exam schedule overlaps with existing exam.

👥 Roles & Authorization Model

📌 Overview

The Examination Management System uses a **Role-Based Access Control (RBAC)** model.

There are **4 main roles**:

1. 🛠️ Admin
2. 🏫 Training Manager
3. 👨‍🏫 Instructor
4. 🎓 Student

Each role has clearly defined permissions to ensure:

- 🔐 Data security
- 🧩 Responsibility separation
- 🚫 Prevention of unauthorized access

- ✅ Controlled system operations

🛠️ 1️⃣ Admin Role

🎯 Purpose

The Admin has full system control and is responsible for system configuration and supervision.

✅ Permissions

- Manage Users (Create / Update / Delete)
- Manage Branches, Tracks, Intakes
- Create Courses
- Assign Instructors
- Create Classes
- Monitor Exams
- View All Reports
- Manage Roles

📊 Database Access

Object Type	Access Level
Tables	Full Access
Views	All Admin Views
Stored Procedures	All Management SPs
Triggers	Indirect (System Controlled)

```sql
-- Testing

-- 1. Switch context to a Admin
EXECUTE AS USER = 'itgate_admin';

-- 2. Try to select from a table (This should work because the admin has full access)
SELECT * FROM dbo.Student;

--- 3. try to select a view from admin view
Select * from v_adm_DatabaseUsageStatistics;
```

99 % No issues found

Results | Messages

	StudentID	ST_name	Email	UserID	BranchID	Track_ID	Intake_ID
1	5	Ahmed Aly	ahmed.aly@iti.edu	12	1	8	2
2	6	Mona Hassan	mona.hassan@iti.edu	13	1	9	2
3	7	Omar Farouk	omar.farouk@iti.edu	14	2	8	2
4	8	Sara Mahmoud	sara.mahmoud@iti.edu	15	2	9	2
5	9	Yassin Ezzat	yassin.ezzat@iti.edu	16	3	8	2
6	10	Nour Zaki	nour.zaki@iti.edu	17	3	9	2
7	11	Hany Salem	hany.salem@iti.edu	18	4	8	2
8	12	Mariam Ali	mariam.ali@iti.edu	19	4	9	2
9	13	Khaled Idris	khaled.idris@iti.edu	20	1	8	2
10	14	Laila Soliman	laila.soliman@iti.edu	21	1	9	2
11	15	Hassan Reda	hassan.reda@iti.edu	22	2	8	2

	StatisticType	Category	TotalCount
1	Exams Per Year	2026	3
2	Student Per Intake	5	1
3	Student Per Intake	2	28
4	Total Answers Stored	All Years	4

🏫 2️⃣ Training Manager Role

🎯 Purpose

Responsible for academic planning and monitoring branch performance.

✅ Permissions

- View branch performance
- View instructors per branch
- View students per track

- Monitor exam results
- View intake reports
- Cannot modify core system structure

📊 Database Access

Object Type	Access Level
Views	Manager Views Only
Stored Procedures	Reporting SPs

```sql
-- Testing

-- 1. Switch context to a Training Manager
EXECUTE AS USER = 'iti_manager';

-- 2. Try to select from a an admin view (This will not work due to fact training manager doesn't have permssion to view admin views)
SELECT * FROM dbo.v_adm_AllInstructor;

--- 3. try to select a view from manager  view  ( will work because it in his premissions )
Select * from v_man_StudentPerformanceRanking;

-- 4. IMPORTANT: Always go back to your Admin account
REVERT;
```

99 % No issues found

Messages

```
Msg 229, Level 14, State 5, Line 26
The SELECT permission was denied on the object 'v_adm_AllInstructor', database 'ExamsSystem_V2_Constraints_SPs_Triggers_views_', schema 'dbo'.

Completion time: 2026-02-20T21:15:29.0504926+02:00
```

```sql
-- 1. Switch context to a Training Manager
EXECUTE AS USER = 'iti_manager';

-- 2. Try to select from a an admin view (This will not work due to fact training manager doesn't have permssion to view admin views)
SELECT * FROM dbo.v_adm_AllInstructor;

--- 3. try to select a view from manager  view  ( will work because it in his premissions )
Select * from v_man_StudentPerformanceRanking;

-- 4. IMPORTANT: Always go back to your Admin account
REVERT;
```

99 % No issues found

Results | Messages

	StudentID	Username	TotalScore	RankPosition
1	5	ahmed.aly	20	1

👨‍🏫 3️⃣ Instructor Role

🎯 Purpose

Responsible for exam creation and student evaluation.

✅ Permissions

- Create Exams
- Add Questions
- Assign Question Degrees
- View Students in Class
- Grade Students
- View Exam Results

❌ Restrictions

- Cannot modify branch or intake
- Cannot manage users

📊 Database Access

Object Type	Access Level
Tables	Limited
Views	Instructor Views
Stored Procedures	Exam & Grading SPs
Triggers	Enforced automatically

```sql
-------Testing Manager

-- Testing

-- 1. Switch context to a Training Manager
EXECUTE AS USER = 'ahmed_instructor';

-- 2. Try to select from a an Manager view (This will not work due to fact Instrctor doesn't have permssion to view admin or training manager views)
Select * from v_man_StudentPerformanceRanking;


--- 3. try to select a view from manager  view  ( will work because it in his premissions )
Select * from v_man_StudentPerformanceRanking;
```

99 % No issues found Ln: 45, Ch: 1 (47 ch

Messages

```
Msg 229, Level 14, State 5, Line 45
The SELECT permission was denied on the object 'v_man_StudentPerformanceRanking', database 'ExamsSystem_V2_Constraints_SPs_Triggers_views_', schema 'dbo'.

Completion time: 2026-02-20T21:18:14.7314977+02:00
```

```sql
-------Testing Manager

-- Testing

-- 1. Switch context to a Training Manager
EXECUTE AS USER = 'ahmed_instructor';

-- 2. Try to select from a an Manager view (This will not work due to fact Instrctor doesn't have permssion to view admin or training manager views)
Select * from v_man_StudentPerformanceRanking;


--- 3. try to select a view from ins  views  ( will work because it in his premissions )
Select * from v_Ins_QuestionBankByInstructor;

-- 4. IMPORTANT: Always go back to your Admin account
REVERT;
```

99 % No issues found

Results Messages

	InstructorID	InstructorName	CourseID	CourseName	QuestionID	Qtext	Qtype	Degree
1	4	itgate_admin	1	SQL Server Fundamentals	1	Which of the following is used to remove all recor...	MCQ	5
2	4	itgate_admin	1	SQL Server Fundamentals	4	Which join returns all rows when there is a match...	MCQ	5
3	4	itgate_admin	1	SQL Server Fundamentals	5	Updated question?	TF	10
4	4	itgate_admin	1	SQL Server Fundamentals	9	Explain the concept of Normalization and its impo...	Text	10
5	4	itgate_admin	1	SQL Server Fundamentals	14	What is SQL?	MCQ	5
6	4	itgate_admin	1	SQL Server Fundamentals	16	What is SQL?	MCQ	5
7	4	itgate_admin	1	SQL Server Fundamentals	18	What is SQL?	MCQ	5
8	6	ahmed_instructor	3	Python for Data Science	2	In Python, which keyword is used to create a fun...	MCQ	5
9	6	ahmed_instructor	3	Python for Data Science	6	Python is a compiled language.	TF	5
10	6	ahmed_instructor	3	Python for Data Science	10	Describe the difference between a list and a dicti...	Text	10

🎓 4️⃣ Student Role

🎯 Purpose

Participates in exams and views personal results.

✅ Permissions

- View Available Exams
- Submit Answers

- View Personal Results
- View Course Grades

❌ Restrictions

- Cannot modify exam structure
- Cannot modify grades

📊 Database Access

Object Type	Access Level
Tables	No Direct Access
Views	Student Views Only
Stored Procedures	Submission & Result SPs
Triggers	Enforced automatically

```sql
-- Testing

-- 1. Switch context to a Student
EXECUTE AS USER = 'omar_student';

-- 2. Try to select from a an ins views (This will not work due to fact student doesn't have permssion to view admin or training manager or ins views)
Select * from v_Ins_QuestionBankByInstructor;


--- 3. try to select a view from ins  views  ( will work because it in his premissions )
Select * from v_Ins_QuestionBankByInstructor;

-- 4. IMPORTANT: Always go back to your Admin account
REVERT;
```

99 % No issues found Ln: 57, Ch: 1

Messages

```
Msg 229, Level 14, State 5, Line 64
The SELECT permission was denied on the object 'v_Ins_QuestionBankByInstructor', database 'ExamsSystem_V2_Constraints_SPs_Triggers_views_', schema 'dbo'.

Completion time: 2026-02-20T21:20:50.2996101+02:00
```

```sql
-- Testing

-- 1. Switch context to a Student
EXECUTE AS USER = 'omar_student';

-- 2. Try to select from a an ins views (This will not work due to fact student doesn't have permssion to view admin or training manager or ins views)
Select * from v_Ins_QuestionBankByInstructor;


--- 3. try to select a view from ins  views  ( will work because it in his premissions )
Select * from v_stu_StudentAnswerReview;

-- 4. IMPORTANT: Always go back to your Admin account
REVERT;
```

99 % No issues found Ln: 68, Ch: 1 (41 chars, 1

Results Messages

	STID	ExamID	QText	IsCorrect	ModelAnswer	StudentAnswer
1	5	2	Which of the following is used to remove all rec...	1	Truncate	TRUNCATE
2	5	2	Which join returns all rows when there is a matc...	1	Full Outer Join	Full Join
3	5	2	Updated question?	0	Review Required / Subjective	True
4	5	2	Updated question?	0	Review Required / Subjective	True
5	5	2	Explain the concept of Normalization and its imp...	1	Review Required / Subjective	Normalization reduces redundancy...